Forecasting Retail Sales at Every Grain: Trees, Foundation Models, Transformers and World Models Compared on Four Public Data Sets

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Abstract. A retailer makes decisions from the same demand at different grains and on different cycles: store orders by item and day, promotions by item and week in blocks of weeks, category reviews week by week, and supplier ranges and terms over a year. It wants one forecast that serves all of them without losing value between plans. This paper asks whether one forecast, built at the finest grain and summed to every coarser grain, can serve these decisions across every kind of line, and which kind of model does that best. It compares gradient-boosted trees (LightGBM), pretrained foundation models (TimesFM, Chronos-Bolt, Timestral), an attention-based model (the Temporal Fusion Transformer) and a world model that carries each item's state forward under the price and promotion plan, on four public data sets (M5, six Dominick's categories, Breakfast at the Frat and Corporación Favorita), with the world model and LightGBM tuned by the same search. Results are read by grain, by the nature of the line, on seen and unseen data, and across retailers. On lines and stores the models were trained on, the tuned world model and LightGBM are close, each ahead on different data sets and grains, and their average is the most accurate choice across grains: it is in the 90% model confidence set at 51 of 60 data-set-and-grain pairs, for the fast lines that carry most sales and for the slow tail. On items kept out of training, the world model loses at most 0.040 in scaled error and forecasts within 11% of actual sales, while LightGBM loses up to 0.311 and forecasts as little as two thirds of actual sales, which pulls the average below the world model alone on all five data sets tested. Settings tuned on one retailer's data carry to another's, and training on other retailers' data improved neither model by more than 0.01. On these data, the rule that follows is one forecast for every item in every store, the average of the tuned world model and tuned trees summed to every grain, with the world model alone for items the models have not been trained on. The rule holds for the data sets studied; anyone adopting the framework should rerun every comparison, including those that lost here, on their own data. The study covers forecasts of 8 weeks and 28 days and the price and promotion levers; whether the models hold over 26- and 52-week plans is left for work with a retailer's own data.

Keywords: retail forecasting; forecast grain; hierarchical evaluation; forecast combination; promotions; new products; world models; foundation models.

1 Introduction

Retail decisions run at different grains and on different clocks. Supplier range, space and cost reviews look at an item's volume across the chain over 26 or 52 weeks, and the terms agreed in them commit allowances for the year ahead. Promotions are planned item by week in blocks of six to twelve weeks, and the promotional and retail allowances that fund them follow the same blocks. Merchants judge their category week by week. Store inventory is ordered item by store, day by day [5]. Each of these decisions reads the same demand at a different grain and over a different span of time.

Since there is no single grain, there is no reason to expect one model to be the most accurate at every grain and every horizon. Retailers do not want a separate model at each grain either. Separate models disagree, and value leaks at each hand-off from a coarse long-range plan to a finer short-range one: the annual plan agreed with a supplier is broken into promotion blocks, weeks and store orders, and where the finer forecasts do not add back to the plan the gap is paid for in stock, markdowns or lost sales. The forecasting literature treats this as coherence across the levels of the product and store hierarchy [26, 27] and across levels of time [22], usually by forecasting each level and reconciling the forecasts. This paper takes the other route retailers use: one forecast made at the finest grain, item by store by period, and summed to every coarser grain, so that the grains add up by construction. The question this paper answers is whether one forecast, built once at the finest grain, can serve the decisions a retailer makes from it, across every kind of line, and which kind of model, or combination of models, does that best.

We test four kinds of model. Gradient-boosted trees are the proven method: LightGBM was behind the leading entries of the M5 competition [1]. Pretrained foundation models (TimesFM, Chronos-Bolt and Timestral)

forecast without training on the retailer's data. The Temporal Fusion Transformer is an attention-based model trained on it. World models, the newest of the four, learn a state for each item and carry it forward one period at a time under the price and promotion plan; ours is shown in Figure 2. The world model and LightGBM were each tuned by the same search on every data set, and the average of forecasts is tested alongside them. The data are four public data sets: M5, six categories of Dominick's, Breakfast at the Frat, and Corporación Favorita.

The study covers the short end of the planning cycle. Forecasts run 8 weeks ahead on the weekly data and 28 days ahead on the daily data, and are scored at every grain from item by store to the chain total, period by period and summed over the whole window as a promotion block. Forecasts of 26 or 52 weeks are not tested: judging them needs a retailer's own plans and range decisions to score against, which no public data set carries, and is left for work with an industry partner. Of the four Ps of retail, the study covers price and promotion, through the plan inputs these data record; product and place are out of scope. Every analysis was written down before it ran, except checks of unexpected results and corrections, which are marked where they appear (Figure 1).

Section 2 sets out the framework, and the results follow it part by part: grain, the nature of the line, seen and unseen data, and transferability, with the broad view last.

Section 3 places the work among retail promotion models, hierarchical evaluation, forecast combination, pretrained forecasters and world models. Sections 4 to 6 give the data, the models and the experimental setup. Section 8 reports the results, Section 10 what was tried and dropped, and Sections 11 to 13 discuss what the results mean for a retailer, their limits, and conclude. The development study is in Supplement B, and the final windows and dropped changes of the first version in Supplement C.

2 Framework

The overall question of this paper is:

> Can one forecast, built once at the finest grain of item by store by period and summed to every coarser grain, serve the decisions a retailer makes from it across every kind of line, and which kind of model, or combination of models, does that best?

It is answered through four research questions, in the order a retailer meets them:

RQ1, grain. At which grains of product, store and time does each model forecast best, period by period and over a whole promotion block, when every grain is summed from the same item-by-store forecast?

RQ2, nature of the line. How does each model do on the fast lines that carry most of the sales, on the long tail of slow lines, and on lines with short history?

RQ3, seen and unseen data. How much accuracy does each model lose on items and stores it was not trained on?

RQ4, transferability. Do a model's settings carry to another retailer, and does training on other retailers' data make it more accurate?

These are the questions retailers ask of a forecast in practice (Section 17). Each question fixes what is measured before any model is compared, and the results are reported in the same order (Section 8). The broad view then brings the four answers together.

RQ1, grain. Does the forecast hold up at every grain a decision is made at, and do the grains add up? Every model forecasts each item in each store for each period, and the forecasts are summed to item, store, category, category by store and the chain total. Each grain is scored period by period, which is how store orders and the category week are judged, and over the whole forecast window as one block, which is how a promotion is judged. Because every grain is a sum of the same item-by-store forecast, the plans built from them add up, and any value lost between grains shows as a difference in accuracy between them.

RQ2, nature of the line. Does the forecast hold up on every kind of line? Grocery sales are concentrated: a short head of fast lines carries most of the units, and a long tail of slow lines sells a few units a week or none. Lines also differ in how much history they have. The fast lines are defined as the items that together make 80% of unit sales over the year before the forecast, the slow lines as the rest, and lines with short history as those whose first sale falls less than 26 weeks before the forecast. Each part is scored on its own series at every grain, so that the total of the fast lines, the slow lines of each store, and so on, are each judged as a planner would judge them. The share of items that makes 80% of sales is reported for each data set.

RQ3, seen and unseen data. Does the forecast hold up on items and stores the model was not trained on? Ranges change and stores open, so part of every forecast is for lines or stores with little or no history in the model.

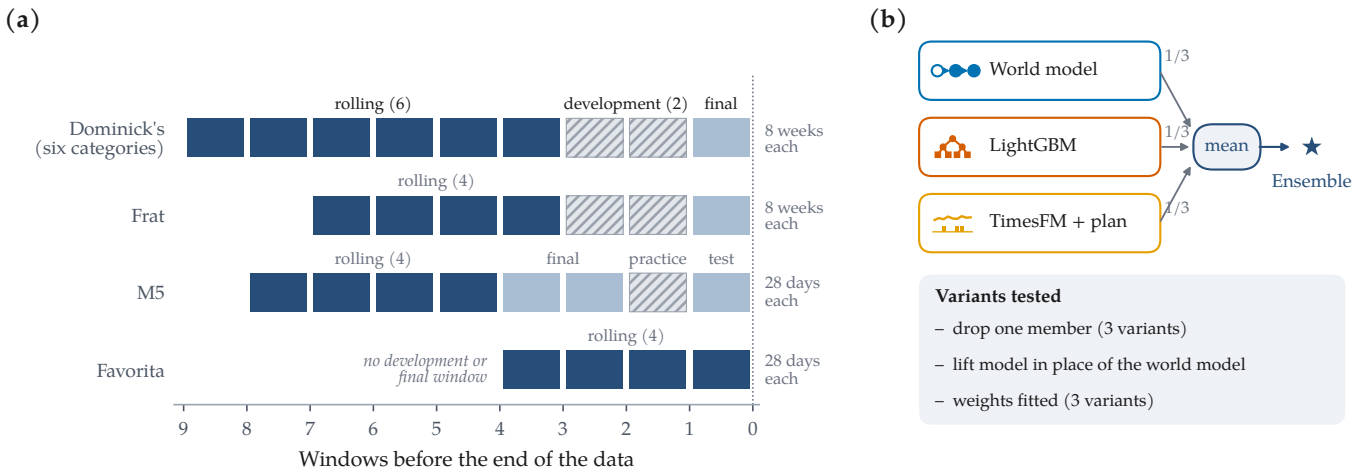


Figure 1: The main results come from rolling windows taken before any window used to choose a setting, and the average tested is fixed in advance. **(a)** Windows by data source, in periods before the end of the data: rolling windows (scored for this revision), development windows (used to choose settings) and final windows (scored once in the first version). Favorita has rolling windows only; nothing was chosen on it. **(b)** The ensemble: the equal-weight mean of the world model, LightGBM and TimesFM with the plan, and the variants of study A that test what each member adds.

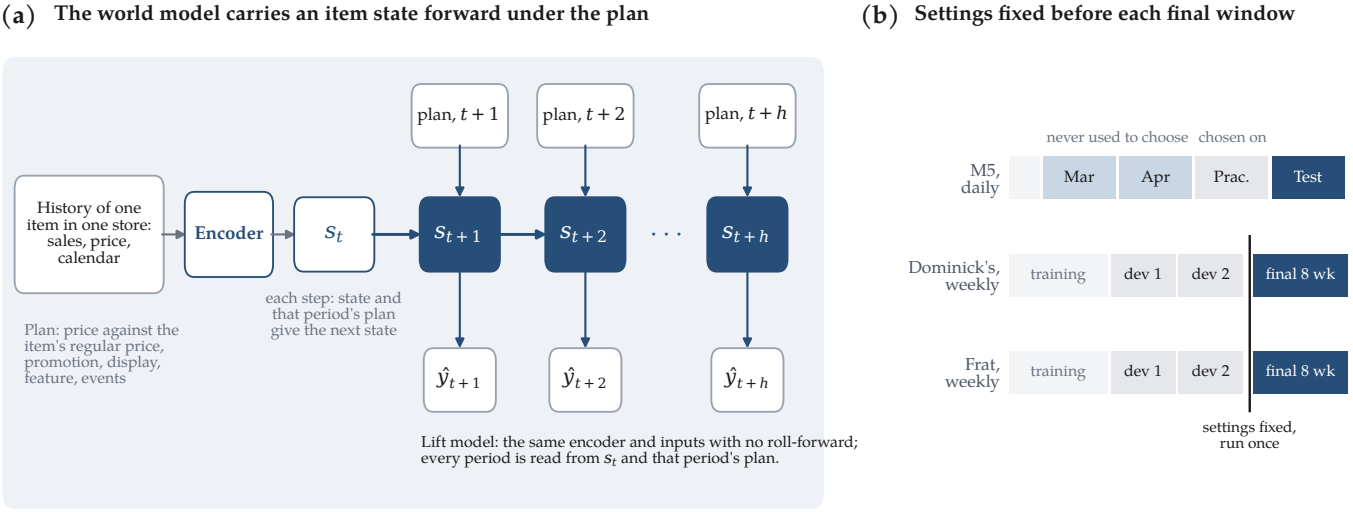


Figure 2: The world model carries each item's state forward under the plan. **(a)** An encoder turns the recent history of one item in one store into a state s_t. A transition step takes the state and one period's plan and gives the state one period later; each period's sales are read from the state as it stands after the plan so far. The lift model shares the encoder and inputs and reads every period from s_t. **(b)** On M5, settings were chosen on the practice window (25 April to 22 May 2016); on Dominick's and Frat, on two development windows before the final 8 weeks.

A fifth of the items, or of the stores, is kept out of training; once trained, each model reads the kept-out series' recent sales and forecasts them. The cost of not having trained on them is the accuracy lost on that fifth against a model trained on everything.

RQ4, transferability. Do the settings carry to another retailer, and does training on other retailers' data help? Settings tuned on M5 are carried unchanged to Favorita and compared with settings tuned on Favorita itself. Each model is also trained on its data set together with the other data sets, and compared with the same model trained on its own data alone.

The broad view. The four parts are then brought together: which model, or combination, gives the best balance across grains, lines and seen and unseen data, and where each kind of model fits in the planning cycle. Findings are reported for each grain and each kind of line, since that is what a retailer needs to choose a model for a decision; the broad view simplifies them and does not replace them.

3 Related work

3.1 Retail forecasting and the promotion plan

Fildes, Ma and Kolassa [5] review retail forecasting research and practice and note that most studies forecast and evaluate at one level of aggregation, usually the item, although retail decisions are spread across levels. Sales of a single item in a single store are intermittent, which led to dedicated methods for intermittent demand [6, 7] and to evaluating forecasts as count distributions [8]. The Tweedie family [9], a compound Poisson–gamma distribution with a point mass at zero, is the loss many M5 entries used, and the loss we use.

Promotions move retail sales more than anything else in the plan. PromoCast forecast promoted-week sales for a chain from the planned event and the item's past response [10]. Van Heerde, Leeflang and Wittink [11] and Macé and Neslin [12] measured the dips in sales before and after a promotion, when shoppers stock up and then buy less. Ma, Fildes and Huang [13] showed that promotional information from within and across categories improves item-level forecasts. Gupta [14] split the promotion bump into brand switching, purchase timing and quantity, and van Heerde, Gupta and Wittink [15] showed that in units only about a third of the bump is switching from other brands. Attraction models make an item's sales the category's sales times its share [16, 17]; our brand-switching step follows that structure. Dominick's Finer Foods ran pricing experiments across its stores [18], and its scanner data are behind much of this work [19].

3.2 Decisions across grains and lines

Retail decisions read the same demand at different grains and over different spans: replenishment item by store by day, promotions item by week, category reviews by week, and range, space and supplier terms over a year or more [5]. Forecasts made separately for each grain need not agree, and the literature restores agreement by reconciliation, across the product and store hierarchy [26, 27], across levels of time [22], and across both together [23]. The alternative studied here, a single forecast at the finest grain summed upward, is coherent by construction and is the way the M5 competition scored its entries [1]; its cost is that one model must serve every grain.

Lines also differ in how much they sell and how much history they have. A small share of items carries most of the units, and the long tail sells a few units a week or none, which led to dedicated methods for intermittent demand [6, 7]. Lines with little or no history are forecast from the histories of similar products: by matching a new product to the life-cycle curves of earlier ones [24], or from product attributes [25]. Global models trained across many series [30] can forecast a series they were not trained on from its recent sales alone, which is how the models in this study meet new items and new stores.

3.3 Hierarchies, combinations and evaluation

Forecasts that must add up across a hierarchy are studied as hierarchical forecasting, from bottom-up and top-down to optimal reconciliation [21, 26, 27]. Our models forecast item by store and their forecasts are summed, as the M5 accuracy competition scored entries, with a weighted, scaled error averaged over twelve levels [1]; reconciliation is tested as a benchmark. The scaling follows Hyndman and Koehler [28], and a companion competition scored forecast distributions [2]. Tashman [29] sets out why a comparison should use several forecast origins. Hansen, Lunde and Nason's model confidence set [73] gives the set of models that cannot be told apart from the best.

Combining forecasts from different models reduces error [36, 37], and an equal-weight average is hard to beat with estimated weights, the forecast combination puzzle, which Smith and Wallis [68] explain by the error in estimating the weights; Wang and co-authors [69] review the field. We fix an equal-weight ensemble in advance and test weights fitted on earlier windows against it. Interval forecasts by calibrating each model on its own past errors follow conformal prediction [70].

3.4 Global models, trees and neural forecasters

Global models, fitted across many series at once, are often more accurate than one model per series [30]. In M5 the leading entries were built on gradient-boosted trees [31, 32]; the winner averaged direct and recursive LightGBM models [4]. Neural global models include N-BEATS [33], DeepAR [34] and the Temporal Fusion Transformer [35], the last two taking known future inputs such as price. Graph neural networks share information between related series [38]; our item-to-item experiments use attention [63] and conditional similarity networks [39].

3.5 Pretrained foundation models

Pretrained time-series models forecast series they were not trained on. TimesFM [40] is a decoder-only transformer pretrained on a large corpus of series; Chronos [42] tokenises values and trains a language model on them, and Chronos-Bolt is its faster direct-quantile version; Lag-Llama [43], Moirai [44] and TimeGPT [45] follow similar lines. TimesFM 2.5 adds

a covariate method that fits a linear model on future inputs and forecasts the remainder with the pretrained model [41]. Timestral 1.0 [72], released by Mistral on 15 October 2026, reads known future inputs inside the model itself and returns 21 quantiles. We test whether any of them, given the plan, matches models trained on it.

3.6 World models and latent prediction

A world model learns a state of its environment and how that state moves under actions, so that the effect of a planned action can be rolled forward before it is taken [46, 47, 48]. LeCun [49] proposed joint-embedding predictive architectures (JEPA), which predict the representation of the future rather than the future itself; I-JEPA [50] and V-JEPA 2 [51] apply the idea to images and video, the latter with an action-conditioned predictor used for planning. Such models guard against a collapsed representation with a slowly updated target encoder [52], a variance and covariance penalty [53], or the isotropic-Gaussian penalty of LeJEPA [54], also used in LeWorldModel [55]. TD-MPC2 [56] rolls a latent state forward under actions; DWM [57] keeps the effect of an action apart from the world's own dynamics, and AD-WM [58] trains world models to tell actions apart for counterfactual planning. Stable-worldmodel [59] evaluates world models under controlled changes, as our perturbation tests do. Closest to us, WorldTS [60] learns covariate-conditioned latent dynamics for forecasting with known future inputs. Our earlier work, M-JEPA [61], compared JEPA fine-tuning recipes with leakage-resistant splits. In retail the actions are the plan. Our world model uses a recurrent transition [62] with layer normalisation [64] on a transformer encoder [63], and is compared with the same model without the transition so that what the carried state adds is measured on its own.

3.7 What this paper adds

Two findings here are not new. That an equal-weight average of different models is hard to beat, and that fitted weights rarely help, is the forecast combination puzzle [68, 69]; that a gradient-boosted tree is a strong single model on retail panels was the lesson of M5 [1]. Hierarchical forecasting asks how to make forecasts at different levels agree, and scores them over all levels together [26, 27, 1]. This paper asks a different question: which model to choose for the grain at which a given decision is made, when the price and promotion plan is known. It adds three things to the work above. It scores each grain separately, ties each to the decision made there, and reports how many models each grain's data can separate, rather than one score averaged over levels; on these data the answer differed between fine and coarse grains. It compares trained plan-aware models with pretrained models given the plan as covariates on the same windows and grains. And it measures, inside the average, what a model that carries a state forward under the plan adds beyond the same network without it.

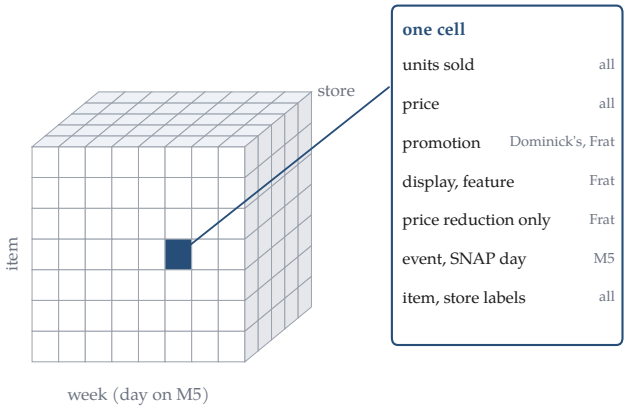
Table 1: Four public data sets, all scored item by store and at every level above it. Rolling windows were added in this revision and precede every window used for development.

Data	Series	Plan inputs	Rolling windows
M5 (daily)	30,490	Price, event days, SNAP days	Four of 28 days, Nov 2015 to Feb 2016
Dominick's (weekly)	Six categories	Price, promotion	Six of 8 weeks per category
Frat (weekly)	3,909	Price, promotion, display, feature, price reduction only	Four of 8 weeks
Favorita (daily)	Top 10 stores	Promotion, holidays and events, paydays	Four of 28 days, the last ending 15 Aug 2017

4 Data

All four data sets are public (Table 1) and share one table shape (Figure 3a). *M5* holds daily Walmart sales of 3,049 items in 10 stores across three states from 2011 to 2016, with prices by week, event days and the days on which SNAP food assistance is paid [1]. *Dominick's Finer Foods* holds weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [19]. We use six categories chosen before any final run (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees). *Breakfast at the Frat* holds weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with the shelf price, base price, and whether the item was on display, in the weekly circular (feature) or on a tag-only price reduction [20]. *Corporación Favorita* holds daily sales of an Ecuadorian grocery chain to 15 August 2017 [71]. It was added in this revision as a data set kept out of development: the series are the 10 stores with the most units sold in 2016 crossed with the items sold in all ten that year, from 1 January 2015; returns are set to zero; the plan is the promotion flag, national holidays and events, and paydays (the 15th and the last day of each month). Favorita records no prices, so its WRMSSE weights are unit weights, not revenue weights. No raw data is redistributed.

(a) One row per item, store and period



(b) The levels a forecast is scored at, and who plans there

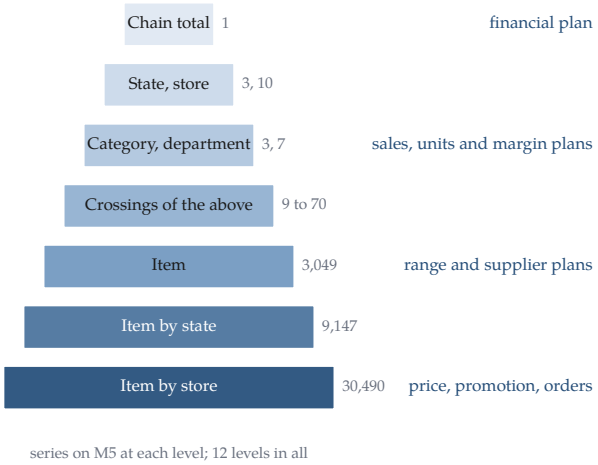


Figure 3: Every data set is one table of item, store and period, scored at every level a retailer plans at. (**a**) Each cell holds the units sold, the price and whatever plan the data set records; a lever a data set does not record is passed to the models as unknown, not as zero. (**b**) The M5 hierarchy with the number of series at each level, and the decisions usually made at each. Dominick's and Frat have six levels: total, category, store, category by store, item and item by store; Favorita has total, store, family, store by family, item and item by store.

Measure. We use the weighted root mean squared scaled error (WRMSSE) of M5 [1] at each level of the hierarchy: 12 levels on M5 (Figure 3b) and six on the others. For a level with series j, history $y_{j,1}, \dots, y_{j,T}$ before the window and forecasts $\hat{y}_{j,t}$ for the H periods after it,

$$\mathrm{RMSSE}_j = \sqrt{\frac{\frac{1}{H}\sum_{t=T+1}^{T+H}(y_{j,t}-\hat{y}_{j,t})^2}{\frac{1}{T-t_j}\sum_{t=t_j+1}^{T}(y_{j,t}-y_{j,t-1})^2}}, \qquad (1)$$

$$\mathrm{WRMSSE} = \textstyle\sum_j w_j \,\mathrm{RMSSE}_j, \qquad (2)$$

where t_j is the series' first period with sales and w_j is its share of the level's dollar sales (units on Favorita) over the last weeks of training. We report item by store, where orders and promotions are decided, the chain total, which a financial plan is set against, and the mean over all levels (the M5 score). Lower is better.

5 Models

Table 2 lists every model. Each trained model's technique, loss and settings are in this section and Table 3; the constraints each works within are in Appendix A.

5.1 World model

Encoder and state. For item i in one store, an encoder (a transformer over the last L periods) reads sales and price, each scaled by the item's recent level ℓ_i, together with the calendar and the recorded levers, and gives a state s_t; the item and store labels enter the readout. The item's level is its average over the history window, or, on Dominick's, over its last 8 weeks.

Table 2: Every forecast compared. Plan: sees the plan for the forecast periods. Trained: fitted on the data set. Timestral is also run from history alone.

Model	Plan	Trained	Technique and loss
Ensemble	yes	yes	Equal-weight mean of world model, LightGBM, TimesFM with the plan
World model	yes	yes	Item state rolled forward under the plan; Tweedie
Lift model	yes	yes	Same network, no roll-forward; Tweedie
LightGBM	yes	yes	Direct tree model per store; Tweedie
Temporal Fusion Transformer	yes	yes	Attention over history and plan; negative binomial
Graph model (M5 final only)	yes	yes	LightGBM's inputs with message passing; Tweedie
TimesFM with the plan	yes	no	TimesFM 2.5 with its covariate method
Timestral with the plan	yes	no	Timestral-28M with known future inputs
TimesFM	no	no	TimesFM 2.5, 512 periods of units
Chronos-Bolt	no	no	Chronos-Bolt base
Seasonal naive	no	no	Sales one season earlier

Transition. A gated recurrent transition f carries the state forward under the plan p_{t+h} for each future period,

$$s_{t+h}(p) = \mathrm{LN}(f(p_{t+h},\, s_{t+h-1}(p))), \quad s_t(p) = s_t, \qquad (3)$$

where LN is layer normalisation (used on M5, Dominick's and Favorita). The plan holds the period's price as a share of the item's regular price (its highest price in the last 12 periods) and the recorded levers.

Readout and loss. Expected sales are a baseline under the "nothing planned" plan p^0 (regular price, no lever) times a lift,

$$\begin{aligned}\log \mu_{i,t+h} = \log \ell_i + b(s_{t+h}(p^0), p^0_{t+h}) \\ + g(s_{t+h}(p), p_{t+h}) - g(s_{t+h}(p^0), p^0_{t+h}),\end{aligned} \tag{4}$$

with b and g small networks that also take the item's labels and the horizon step, and the log forecast bounded to $[-10, 10]$. The lift is exactly zero when nothing is planned, and the baseline does not move when the plan changes; both are checked by tests in the repository. Training minimises a Tweedie loss on sales relative to ℓ_i, with batches of item, store and date drawn at random.

Brand switching (Dominick's). In a store and week where several items are promoted, much of each item's lift is taken from the others. Following attraction models [16, 17], the world model fits on the last 8 training weeks one number per run, the share α of the summed lift that the category total keeps, and rescales its item forecasts in each store and week so that the category total is the baseline total plus α times the summed lift: with baselines b_i, lifts λ_i and forecasts $\hat{y}_i = b_i + \lambda_i$ for the items i of one store and week,

$$\hat{y}'_i = \hat{y}_i \, \frac{\sum_k b_k + \alpha \sum_k \lambda_k}{\sum_k \hat{y}_k}, \tag{5}$$

so each item keeps its share of the category forecast. With Λ_g the summed lift and G_g the actual minus baseline sales of store-week group g in the last 8 training weeks, α is the least-squares slope $\sum_g \Lambda_g G_g / \sum_g \Lambda_g^2$, kept between 0 and 1. α is a forecast-implied split of lift, not an identified estimate of where promoted sales come from, because promotions are not assigned at random. On Dominick's three fits with different seeds are averaged in each run.

What the state holds. A probe on the frozen world model shows that the carried state takes in the plan (Figure 4): the day's discount can be read back from the rolled state with R^2 0.58 to 0.83 across three M5 windows, against -0.01 to 0.02 from the state before the window, and still at 0.46 to 0.59 seven days later. That the plan is in the state does not show that the forecast uses it well, which §9.5 tests.

5.2 Lift model

The same model without the roll-forward: $s_{t+h}(p) = s_t$ for every h. On M5, Dominick's and Favorita the two share encoder, readout, size, inputs, training steps and regularisation, so the difference between them is the carried state; it is the transition ablation. On Frat the lift model's settings, carried over from the first version, add readout dropout [65] of 0.2 and decoupled weight decay [66] of 0.01 that the world model does not use; study G reruns it with the world model's settings (§9.3).

Table 3: The world model as run in the final tests. Read from the run manifests in `results/promoted/`. Periods are days on M5 and weeks on Dominick's and Frat.

Setting	M5	Dominick's	Frat
Periods of history read	112	26	26
Periods of training data	730	156	90
State size	128	64	64
Encoder layers	3	2	2
Attention heads	4	4	4
Training steps	30000	6000	3000
Batch size	1024	512	256
Learning rate	0.001	0.001	0.001
Normalised roll-forward	yes	yes	no
Level from the last 8 weeks	no	yes	no
Brand switching	no	yes	no
Fits averaged per run	1	3	1

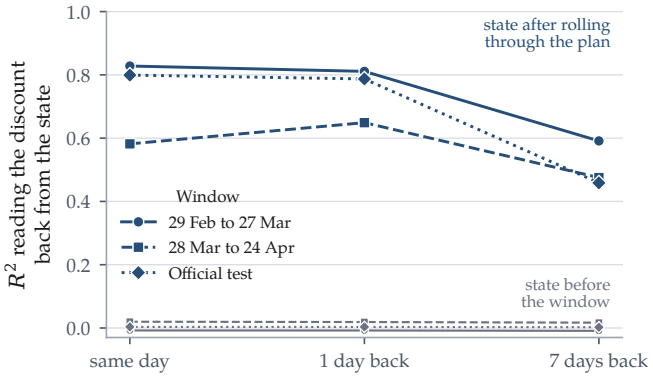


Figure 4: The rolled state holds the plan, and still holds most of it a week later. R^2 of a ridge read-out of the day's discount from the frozen world model's state, fitted on half of 6,000 item-store pairs and scored on the other half. Blue: the state after rolling through the window's plan. Grey: the state before the window. Seed 1 in three M5 windows.

5.3 Training losses

The loss used. The world model and the lift model are trained on sales relative to the item's level, $z = y/\ell_i$, with the Tweedie deviance for a mean μ and power $p = 1.5$,

$$L_p(z, \mu) = -\frac{z\,\mu^{1-p}}{1-p} + \frac{\mu^{2-p}}{2-p}, \tag{6}$$

the negative log-likelihood, up to terms free of μ, of a compound Poisson–gamma distribution with a point

mass at zero [9]. Because the loss is on z, every item counts alike whatever its volume. LightGBM uses the same deviance with $p = 1.1$ on units, so large items count for more, and the Temporal Fusion Transformer a negative binomial likelihood on units.

The changes tried. Each change below adds to or replaces L_p for one batch B of item, store and period examples; Table 4 scores every one against the configuration it changed. None became a default.

- *Negative binomial* on units, $-\log \mathrm{NB}(y;\, \mu\ell_i, r)$ with the spread r learnt.
- *Totals term*: the loss adds $\lambda_T \,\mathrm{mean}_t[(\sum_{b\in B_t} \hat{y}_b - \sum_{b\in B_t} y_b)/\max(1, \sum_{b\in B_t} y_b)]^2$, the relative gap between summed forecasts and sales over the examples B_t sharing a date, so that a small bias in the same direction on many items is penalised ($\lambda_T = 1$).
- *Volume and dollar weights*: each example's loss times ℓ_i^γ, or $(\ell_i\bar{\pi}_i)^\gamma$ with $\bar{\pi}_i$ the item's usual price, normalised to mean one in the batch, with $\gamma = 0.9$ and $p = 1.1$, which makes the loss close to LightGBM's on units or dollars.
- *Peak weight*: each period's loss times $1 + \kappa \min(|z_t - 1|, 5)$, normalised to mean one, so that peaks and dips count for more ($\kappa = 0.5$, with the dollar weight above).
- *After-promotion weight*: each period with no lever on that follows a period with one counts $1 + w$ times, normalised to mean one ($w = 2$ or 5).
- *Latent term*: from the state now and the plan, the transition predicts the state the encoder gives once those periods are history, against a slowly updated copy of the encoder; the loss adds $\lambda_Z \sum_h \delta^h \|s_{t+h}(p) - \bar{s}_{t+h}\|^2$ with $\delta \le 1$ discounting far periods, and optionally a variance and covariance penalty on the state [53]. It was also used alone to pretrain the encoder, which was then frozen, trained at a lower rate or trained fully on the forecast loss.

5.4 Ensemble

The equal-weight mean of the forecasts of the world model, LightGBM and TimesFM with the plan for each item, store and period, fixed on 13 October 2026 before any revision result, with no weights fitted. Its intervals come from its own past errors like any other model's (§6).

5.5 Comparison models

LightGBM [31]: one direct model per store, Tweedie loss, sales at lags no shorter than the horizon, rolling averages and spread of sales, price against the item's highest recent price, calendar and plan inputs. *Temporal Fusion Transformer* [35]: one model per data set on Dominick's and Frat, fitted on the last 156 weeks with the plan as known future inputs and a negative binomial likelihood, 1,500 steps; it was not run on M5, where its training cost was beyond the compute available, nor on Favorita, whose model list was fixed without it. Chronos-Bolt and TimesFM from history alone were not run on Favorita either. *Graph model*: LightGBM's inputs with one round of GraphSAGE-style message passing [38] between the same item in other stores and items of the same department in the same store; run on the M5 final windows only. *TimesFM* [40, 41]: the 200M-parameter TimesFM 2.5 checkpoint given each series' last 512 periods; we use its mean output, because its median forecasts 27% to 31% below M5 totals (§C.1). *TimesFM with the plan*: the same model through its covariate method, a linear model on the plan inputs fitted over groups of 2,048 series, then TimesFM on what that model leaves. *Chronos-Bolt* [42]: the base checkpoint, zero-shot, the mean of its quantiles. *Timestral* [72]: Timestral-28M (code commit a534bdb), zero-shot, with the plan as known future inputs (price relative to the context's highest price, other levers as given) and from history alone; the point forecast is the mean of its 21 quantiles, each weighted by the share of probability nearest it. *Seasonal naive*: sales one season earlier.

6 Experimental setup

Windows. On Dominick's and Frat the last 8 weeks of each data set were the final window of the first version, touched once, and settings were chosen on the two 8-week windows before it. On M5 settings were chosen on the practice window (25 April to 22 May 2016), and the final windows were the official test month and, after it, the two months before the practice window. For this revision each data set gained rolling windows taken before the development windows, so none was used for any choice: six 8-week windows on each Dominick's category, four on Frat, and four 28-day windows on M5 (November 2015 to February 2016), each forecast from everything earlier (Figure 1a). Favorita has four 28-day rolling windows, the last ending on its last day, and no other.

Rules fixed in advance. The revision plan, including every window, model, test and pass rule below, was written down on 13 October 2026 before any revision

Table 4: No change to the training loss became a default: each was lower in error than the configuration it changed in at most half its comparisons. Each row is one change to the Tweedie loss of the world model or lift model, scored on the development windows against the configuration with the same data, windows and every other setting. WRMSSE: overall (mean of the hierarchy levels), averaged over the seeds and windows both ran and over the data sets in the row. Lower: the number of data sets (or category and model pairs) in which the change had the lower error. The weight after a promotion comes from study E, where the world model was rerun in the same job. Rows marked not run were configured but have no full-data run in the records. Built by `scripts/loss_ledger.py`.

Change to the loss	Tested on	With it	Without	Result
Negative binomial likelihood	Dominick's, Frat	0.537	0.541	Lower in 1 of 2
Totals term	Dominick's, Frat	0.655	0.583	Lower in 0 of 3
Unit-volume weight and Tweedie power 1.1	Dominick's	–	–	Not run on full data
Dollar-volume weight and Tweedie power 1.1	Dominick's	–	–	Not run on full data
Peak weight, dollar weight and Tweedie power 1.1	Dominick's	0.853	0.856	Lower in 5 of 12
Weight 3 on the periods after a promotion	Dominick's	0.834	0.832	Lower in 2 of 6
Weight 6 on the periods after a promotion	Dominick's	0.810	0.832	Lower in 2 of 6
Latent term, weight 0.1	Frat	0.411	0.406	Lower in 0 of 1
Latent term, weight 0.5	Frat	0.414	0.407	Lower in 0 of 2
Latent term, weight 2	Frat	0.410	0.406	Lower in 0 of 1
Latent term with a spread penalty and discount	Dominick's, Frat	0.567	0.549	Lower in 0 of 4
Latent pretraining, encoder frozen afterwards	Frat	0.428	0.406	Lower in 0 of 1
Latent pretraining, encoder full afterwards	Frat	0.407	0.406	Lower in 0 of 1
Latent pretraining, encoder at a lower rate afterwards	Frat	0.411	0.407	Lower in 0 of 3
Latent pretraining, then forecast with the latent term	Frat	0.415	0.406	Lower in 0 of 1
Latent pretraining, then forecast with a totals term	Dominick's, Frat	0.729	0.667	Lower in 0 of 1

run, and every change to it is a dated amendment; all but the interval fix below were made before the results they affect were read. Every model runs in every rolling window with the settings of its final configuration, unchanged. Trained models are one fit in the rolling windows: seed 1 where the configuration names a seed (M5, Frat, LightGBM everywhere), and seed 0 for the world model, lift model and TFT on Dominick's, whose configurations name none and which, before a fix on 16 October, did not pass the run's seed to the model (§10). The final windows keep their three-seed results. No new result changes a setting of a model already reported, and every model, window and test is reported.

Comparing two models. For models A and B, window k and a level, let $d_k = \text{WRMSSE}_k(A) - \text{WRMSSE}_k(B)$. Within a window, d_k gets a 95% interval from 1,000 resamples of items: an item drawn twice counts twice in every series and aggregate it feeds, with the weights recomputed. Across the K windows, the mean $\bar{d} = \frac{1}{K}\sum_k d_k$ gets a 95% interval from the 2.5% and 97.5% quantiles of $\bar{d}^*$ over 5,000 resamples of windows with replacement. A model is said to beat another only when that interval excludes zero; when it does not, we say no difference was detected, which is not a finding that the two are equal. For each data set and level we report the 90% model confidence set [73]: starting from all models $\mathcal{M}$, the range statistic $T = \max_{a,b\in\mathcal{M}} |\bar{d}_{ab}|/\hat{\sigma}_{ab}$, with $\hat{\sigma}_{ab}$ from the bootstrap over windows, is compared with its bootstrap distribution under equal expected loss (5,000 resamples); while the p-value is below 0.10 the model whose standardised losses against the others are largest is removed, and the models left form the set. With four to six windows these tests have little power, and adjacent windows are dependent: each is forecast from all data before it, so they share most of their training history, and an error tied to a season or a stretch of trading can recur in neighbouring windows. A bootstrap that resamples whole windows treats them as exchangeable and so understates that dependence; we report the number of windows in which one model beats another next to the interval where it matters (§9.4).

Intervals. Each model's interval forecasts come from its own errors in the earlier rolling windows of the same data set, a rolling form of conformal calibration [70]. With $e_{j,h} = (y_{j,t+h} - \hat{y}_{j,t+h})/\sigma_j$ the scaled errors of the earlier windows at horizon h (σ_j the series' one-step scale), the interval at level $1-\alpha$ is $[\hat{y}_{j,t+h} + \sigma_j q_{\alpha/2,h},\ \hat{y}_{j,t+h} + \sigma_j q_{1-\alpha/2,h}]$, with $q_{\cdot,h}$ the empirical quantiles of the pooled $e_{\cdot,h}$. Where a level has fewer than 50 earlier errors for each horizon, all horizons are pooled into one set of quantiles, and interval ends are rounded outwards to whole units except on Favorita;

this fix was added on 16 October after the first intervals covered far less than planned at the totals, and both versions are reported. The first rolling window has no earlier window and is not scored. We report the scaled pinball loss averaged over the quantiles, weighted as in the M5 uncertainty competition [2], and coverage at 50%, 80% and 95%. Models that give quantiles of their own (Chronos-Bolt, TFT) are also scored on them, over every window.

Decisions. Three simulations turn forecasts into costs. *Store orders*: for each item-store-period, order the model's interval quantile at a critical ratio r of 0.5, 0.8 or 0.95; each unit over costs 1 and each unit short $r/(1-r)$. *Category plan*: for category-by-store nodes n with sales $y_{n,t}$, forecast $\hat{y}_{n,t}$ and average price $\bar{\pi}_n$ before the window,

$$C = \frac{\sum_n \bar{\pi}_n \sum_t |y_{n,t} - \hat{y}_{n,t}|}{\sum_n \bar{\pi}_n \sum_t y_{n,t}}, \tag{7}$$

the absolute error in dollars over sales in dollars; study G checks it against the first half of the window, the error of the whole window's total, units in place of dollars and equal node weights. *Promotion decisions*: in the generated store worlds of the companion paper [67], where each promotion's true lift is known, promote when the predicted lift times margin exceeds a fixed cost; regret is the value lost against deciding with the true lift. The first two are reported against seasonal naive. All are simulations under stated costs, not business results.

Reconciliation. Base forecasts at every node come from AutoETS fitted on the history before each window (seasonal at the aggregate nodes, non-seasonal item by store). They are reconciled bottom-up, top-down by historical proportions, and by the MinT family [27]. With S the summing matrix (nodes by item-store series) and $\hat{\mathbf{y}}$ the base forecasts at every node, the reconciled item-store forecasts are

$$\tilde{\mathbf{b}} = (S^\top W^{-1} S)^{-1} S^\top W^{-1} \hat{\mathbf{y}}, \tag{8}$$

with $W = I$ (ordinary least squares), $W = \mathrm{diag}(S\mathbf{1})$ (structural weights), W the diagonal of in-sample error variances, or $W = \lambda \mathrm{diag}(\hat{\Sigma}) + (1-\lambda)\hat{\Sigma}$ with the Schäfer–Strimmer intensity λ (MinT-shrink, where the hierarchy has at most 6,000 nodes: Frat); nodes with no in-sample error are given the smallest variance of the others, a fix made after the first run failed (§9.9). Each model's own item-by-store forecasts are also combined with ETS above them by weighted least squares.

Ranking. Accuracy says how close a forecast is, not whether it puts items in the right order. For every window we report the Spearman correlation, across item-store series that sold in the window before, between the forecast change (horizon total over the previous horizon's total) and the actual change; the share of the tenth of nodes forecast to grow most that are in the actual top tenth (0.10 by chance); the Spearman correlation across promoted periods between forecast and actual units over the series' unpromoted level; and bias, forecast over actual. These were written down before they ran and have no pass rule.

Studies added after the rolling results. The rolling windows put the ensemble first, so studies were added, each written down before it ran and reported as added after the main results: *A*, what each member adds to the ensemble and whether fitted weights beat equal ones; *B*, input perturbation on the saved models; *C* and *E*, changes aimed at the forecast after a promotion; *D*, Favorita; *F*, Timestral, with the ranking check alongside; and *G*, asked for by a reader of the first revision: paired uncertainty and sensitivity for the category-plan cost, the Frat lift model rerun with the world model's regularisation, and a diagnosis of the MinT failure. Study G found an error in the category-plan cost and a numerical failure in MinT-shrink; both are corrected and the first results kept for the record. Three tests on the saved world models complete the set: shuffling the timing of the plan within each series, adding one promotion and price cut and following its effect through later periods, and a ridge read-out from the rolled state to sales after the window.

7 Searching the settings of both models

Both models were tuned with the same procedure on every data set. On the windows the searches scored, tuning improved LightGBM on all nine data sets and the world model on six, and the gains were mostly small next to the differences between data sets. The held-out window agreed with the search windows only weakly, so the paper judges the models on later windows the searches never saw. Table 5 gives every search from its starting settings to its final settings, and Figure 5 shows every run in the order it was made.

7.1 The procedure

The procedure is a fixed loop, written down before it ran, that changes one setting at a time. It starts from the settings the first version of this paper used. Each change on a menu fixed in advance is run once, on one seed, on top of the best settings so far. Every change that beats the best settings on that seed is run on two more seeds. The best of those is kept if its mean over three seeds beats the best settings' mean by at least the keep bar, which is the spread of the three seeds of the starting settings.

The loop then starts a new round from the new best and stops when a round keeps nothing. A change cannot be kept because one seed was lucky, and the bar is set by the model's own noise on that data set before any change is tried.

The windows the loop scores lie before every window this paper reports. The latest of them is held out: it is scored for every three-seed run and never used to decide. The menus differ between the models, because the models have different settings, but each covers what a practitioner would try first: for the world model the loss weights, the roll-forward, training length, learning rate and batch, weight averaging and the latent checks of Section 5; for LightGBM the number of leaves, the learning rate and number of rounds, the minimum data in a leaf, the objective, feature sampling and the lag windows.

After its loop each world model had two more rounds of one change each. The first tried four count likelihoods that treat weeks with no sales on their own: the negative binomial, the same with a spread for each item, the zero-inflated negative binomial and the hurdle negative binomial. The second tried JEPA pretraining of the encoder with the forecast head fitted afterwards at a lower rate, at two lengths of pretraining. Both used the keep rule above. LightGBM has no counterpart to either, so it had none. Pretraining used only each data set's own history, so the two models still saw the same data.

Each model on each data set was searched as a five-copy average, the mean forecast of five fits with different seeds, except on M5, where both models were single fits. Section 7.4 shows why the average matters.

7.2 What the searches kept

The searches made 1,020 runs, 463 of the world model and 557 of LightGBM, and kept 22 changes, 9 for the world model and 13 for LightGBM. LightGBM kept at least one change on every data set and the world model kept nothing on three of the six Dominick's categories. No single change helped the world model everywhere (Figure 6). Weighting the loss toward high-volume series made the first seed worse on most data sets. The seven-step latent check, kept on Favorita, was among the changes that moved Favorita most and did little elsewhere.

On the search windows the world model improved on six of the nine data sets and LightGBM on all nine (Figure 9). The largest gain of either model was LightGBM on Favorita, from 0.922 to 0.580. The world model's largest was also Favorita, from 0.547 to 0.483.

7.3 Did the held-out window agree?

Only partly. Across every three-seed run, the change on the search windows and the change on the held-out window are correlated at 0.29 for the world model and 0.05 for LightGBM (Figure 7). A change that helps on the search windows is a weak guide to the next window. The world model's final settings were worse than its starting settings on the held-out window on two data sets, Frat (0.591 to 0.608) and snacks (0.568 to 0.608), and LightGBM's on five, most on soft drinks (0.435 to 0.507). This is why the paper reports the final settings on windows that the searches never saw, in Section 8, and not the search scores.

7.4 Seed noise, one copy against five

A single world-model fit is noisy enough to hide most of the changes the loop tries. On the six Dominick's categories the three seeds of the starting settings differ from their mean by a standard deviation of 6.3% for a single fit and 1.8% for a five-copy average (Figure 8). The single-fit search on cheese had a bar of 0.040 and kept nothing; with the average the bars on the Dominick's categories were between 0.004 and 0.034. LightGBM's seeds differ much less, and its bars were between 0.002 and 0.010.

8 Results

The results follow the four research questions. Every number comes from the tuned world model and the tuned LightGBM, each set by the same search on each data set (Section 7), scored on the paper's windows. The average is the equal-weight mean of the two forecasts.

8.1 RQ1: grain

Use the average at every grain. Across the nine data sets and the three views of Table 6 (store grain and above by period, item grain by period, and the whole window as one block), the average has the lowest error in 22 of 27 cases and is never the least accurate of the three. The world model is the least accurate in 16 cases and LightGBM in 11.

The paired tests give the same answer. They compare two forecasts level by level and window by window, and count a difference only when it holds after resampling items. Against the world model, the average is clearly more accurate in 128 of 312 comparisons and clearly less accurate in 27; against LightGBM, 124 and 21. At item grain it is clearly less accurate than either model in at most 8 of 100 comparisons. Across data sets and levels it is in the 90% model confidence set in 51 of 60 cases, against 37 for LightGBM and 35 for the world model.

Table 5: Every search, from starting to final settings. Scores are WRMSSE (lower is better), the mean of three seeds on the search windows and one score on the held-out window, which never decided anything. The keep bar is the seed spread of the starting settings; a change was kept only when its three-seed mean beat the best so far by at least the bar. Tries counts every run, first-seed screens and three-seed confirmations.

Data	Model	Tries	Keep bar	Starting		Final		Kept, in order
				Search	Held-out	Search	Held-out	
M5	World model	59	0.013	0.570	0.550	0.533	0.534	feedback, hurdle nb item spread
	LightGBM	51	0.003	0.603	0.553	0.556	0.522	num leaves quarter
Favorita	World model	51	0.028	0.547	0.600	0.483	0.560	latent overshooting
	LightGBM	99	0.009	0.922	0.576	0.580	0.573	min data in leaf 4x, objective poisson, learning rate half
Frat	World model	53	0.028	0.588	0.591	0.547	0.608	batch half
	LightGBM	52	0.003	0.501	0.546	0.489	0.546	learning rate double
Analgesics	World model	35	0.004	0.509	0.619	0.509	0.619	nothing
	LightGBM	52	0.010	0.548	0.757	0.528	0.764	num leaves quarter
Bottled juice	World model	30	0.028	0.587	0.637	0.587	0.637	nothing
	LightGBM	67	0.008	0.708	0.582	0.674	0.593	objective poisson, feature fraction 0.8
Cheese	World model	33	0.034	0.392	0.714	0.392	0.714	nothing
	LightGBM	49	0.002	0.352	0.833	0.346	0.839	longer windows
Frozen entrees	World model	95	0.007	0.246	0.251	0.216	0.241	total weight 0.1, batch half
	LightGBM	53	0.010	0.222	0.183	0.181	0.195	num leaves 4x
Soft drinks	World model	65	0.020	0.594	0.579	0.572	0.522	lr double
	LightGBM	80	0.004	0.554	0.435	0.532	0.507	num leaves quarter, max bin 255
Snacks	World model	42	0.018	0.439	0.568	0.400	0.608	negative binomial item spread, jepa hybrid long
	LightGBM	54	0.006	0.358	0.425	0.351	0.424	rounds double

Grain does not decide between the two models on their own. The world model is clearly more accurate than LightGBM in 40 comparisons at store grain and above and clearly less accurate in 45; at item grain the counts are 24 and 30. Neither is ahead at either grain across the data sets, so a retailer that picks one model for a grain is picking by data set, not by grain. The average removes that choice.

On the whole promotion block, the average is again the safe choice. It is lowest on five of nine data sets and second on the other four. LightGBM is lowest on analgesics, bottled juice and snacks, but drifts in total where the world model does not: its block error on M5 is 0.953, against 0.598 for the world model and 0.589 for the average, and the same pattern shows on Favorita and soft drinks.

8.2 RQ2: nature of the line

Sales are concentrated, and the average holds up on both the fast lines and the slow tail; where history is short, the world model is ahead of LightGBM. On the six Dominick's categories, 11 to 21% of the items make 80% of unit sales over the year before the forecast; on M5, Frat and Favorita the head is flatter, at 37 to 42% (Table 7). On the fast lines the average has the lowest error on eight of nine data sets, and on the slow lines on eight; on snacks LightGBM is lowest on the fast lines, and on cheese the world model is lowest on the slow lines.

On series whose first sale falls less than 26 weeks before the forecast, the world model's error is lower than LightGBM's on eight of nine data sets, Frat being the exception: for example 0.539 against 0.667 on Favorita and 0.866 against 1.024 on M5. These series are 11 to 21% of the series on the Dominick's categories, where items enter and leave the range over the years, and under 2% on M5, Frat and Favorita.

8.3 RQ3: seen and unseen data

On items the models were not trained on, use the world model alone; on stores they were not trained on, the average serves. With a fifth of the items kept out of training, the world model's error on them rises by 0.008 to 0.040, against 0.058 to 0.311 for LightGBM (Table 8). LightGBM under-forecasts these items, at 66 to 93% of actual sales, and that pulls the average down to 78 to 96%; the world model forecasts 89 to 104%. On the kept-

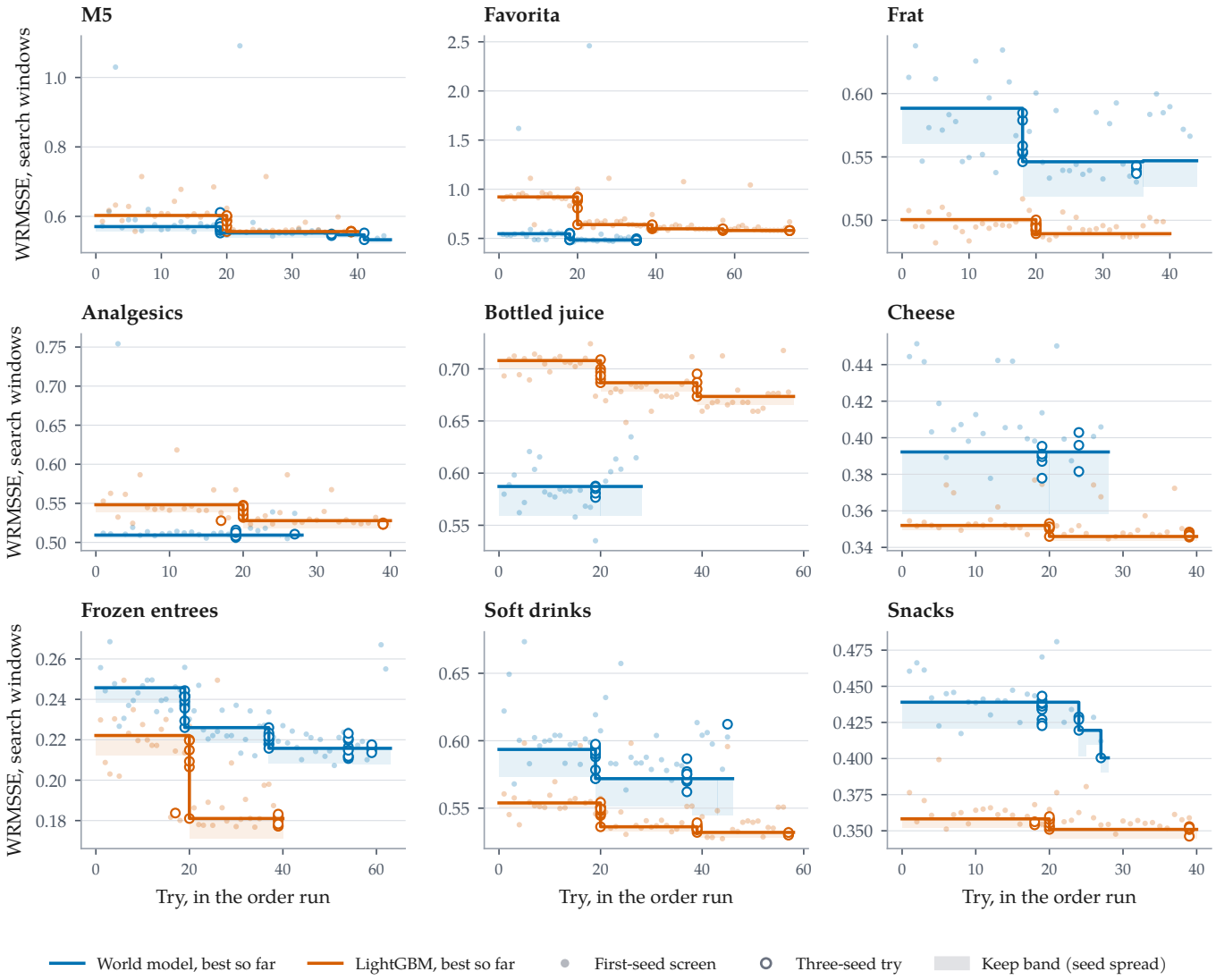


Figure 5: Every run of every search, in the order it was made. Faint dots are first-seed screens, open circles are three-seed confirmations, and the step line is the best settings so far. The band under the line is the keep bar: a change had to land below it to be kept. The world model's line runs on into its likelihood and JEPA rounds after its own loop ends.

out items the world model alone is clearly more accurate than the average on five of five data sets tested.

With a fifth of the stores kept out, the world model is clearly more accurate than LightGBM on four of five data sets and level on M5. Against the average it is clearly more accurate only on Frat; the average is clearly more accurate on cheese, and the two are level on snacks, M5 and Favorita.

8.4 RQ4: transferability

The settings carry to another retailer, and training on other retailers' data does not help either model. M5's tuned settings carried unchanged to Favorita give 0.447 overall for the world model, against 0.428 with Favorita's own settings, and 0.450 for LightGBM, against 0.448 (Table 9); at item by store the difference is 0.002 for the world model and 0.001 for LightGBM. Training each model on its data set together with other data sets lowers the error by no more than 0.007 overall on any data set: by 0.005 for the world model on analgesics and by 0.007 for LightGBM on frozen entrees. Elsewhere it leaves the error unchanged or raises it, by as much as 0.141 for the world model on soft drinks and 0.235 for LightGBM on analgesics.

8.5 The broad view

Brought together, the four answers give one rule. **Forecast every item in every store as the equal-weight average of a tuned world model and tuned gradient-boosted trees, and sum it to every grain; for an item the models have not been trained on, use the world model alone until they are retrained with it.** On the data sets

Table 6: RQ1, grain. WRMSSE of the tuned world model (WM), the tuned LightGBM (LGB) and their equal-weight average (Avg), averaged over the levels at store grain and above, over the item levels, and over every level for the whole forecast window summed as one block; mean over the paper's windows. Shading marks the lowest of the three: world model, LightGBM, average.

	Store and above			Item levels			Whole block		
Data set	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg
Analgesics	0.688	0.698	**0.678**	0.710	0.705	**0.695**	0.523	**0.458**	0.472
Bottled juice	0.569	0.566	**0.539**	0.581	0.596	**0.563**	0.545	**0.404**	0.445
Cheese	0.381	0.471	**0.377**	0.623	0.594	**0.578**	0.445	0.425	**0.387**
Frozen entrees	0.339	0.327	**0.320**	0.375	0.350	**0.344**	0.353	0.330	**0.318**
Soft drinks	0.822	0.805	**0.754**	0.589	0.555	**0.547**	0.576	0.632	**0.470**
Snacks	1.241	**1.040**	1.071	0.545	0.516	**0.508**	0.893	**0.633**	0.688
Frat	0.471	0.479	**0.449**	0.528	0.524	**0.508**	**0.345**	0.406	0.364
M5	0.660	0.634	**0.591**	0.859	0.886	**0.856**	0.598	0.953	**0.589**
Favorita	0.314	0.325	**0.278**	0.655	0.696	**0.653**	0.482	0.565	**0.412**

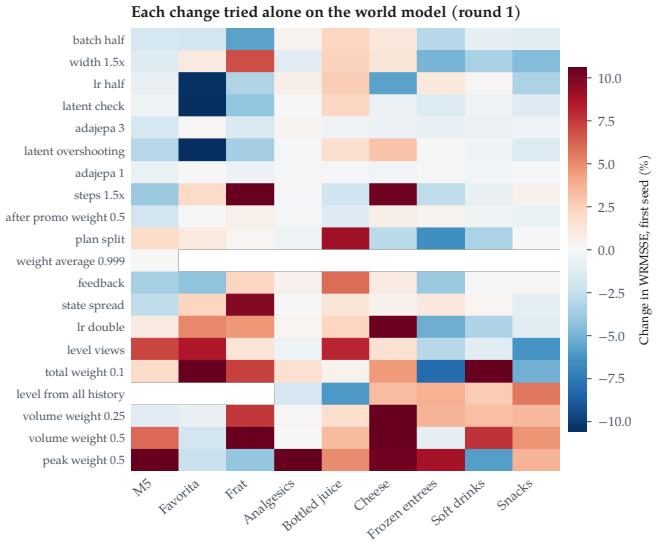


Figure 6: Each change tried alone on the world model in the first round, against that data set's starting settings on the first seed. Blue is better. A blank cell is a change that was not on that data set's menu.

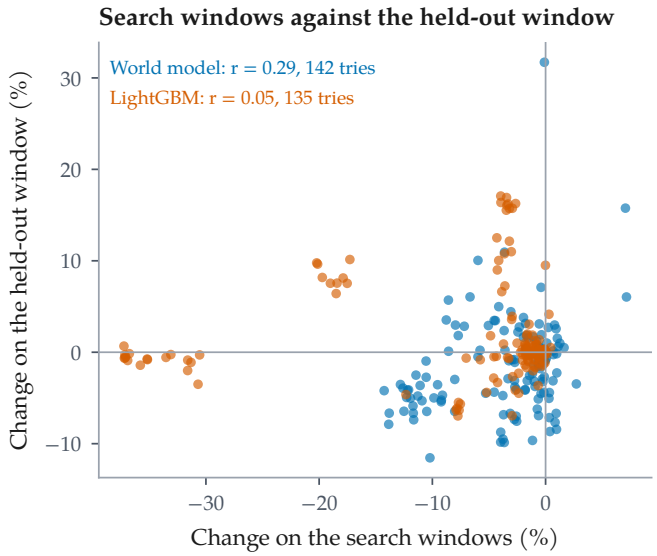


Figure 7: Each three-seed run's change on the search windows against its change on the held-out window, both against the starting settings of its search.

studied, the average is the most accurate choice at store grain and above on eight of nine data sets, at the item levels on all nine, and on the fast lines and the slow tail on eight of nine. The two models on their own are close on items they were trained on; they part company on items they were not, where LightGBM under-forecasts and the world model does not. The trigger for the exception is one a merchant always knows: whether the item is new to the range. Settings carry from one retailer's data to another's, and training on other retailers' data does not

Table 7: RQ2, nature of the line. Share of items that make 80% of unit sales over the year before the forecast, and WRMSSE averaged over every level, by period, on the fast lines (those items), the slow lines (the other items) and the series with short history (first sale less than 26 weeks or 182 days before the forecast). WM, LGB and Avg as in Table 6. Shading marks the lowest of the three: world model, LightGBM, average.

	Items making	Fast lines			Slow lines			Short history		
Data set	80% of units	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg
Analgesics	17%	0.788	0.783	**0.772**	0.714	0.726	**0.707**	**1.348**	1.549	1.425
Bottled juice	21%	0.609	0.621	**0.584**	0.448	0.450	**0.432**	**1.368**	1.480	1.390
Cheese	17%	0.474	0.493	**0.434**	**0.458**	0.544	0.478	**1.309**	1.534	1.389
Frozen entrees	19%	0.331	0.317	**0.311**	0.424	0.367	**0.361**	1.522	1.546	**1.386**
Soft drinks	11%	0.657	0.627	**0.600**	0.608	0.654	**0.596**	1.523	1.537	**1.475**
Snacks	17%	0.689	**0.494**	0.534	1.388	1.349	**1.339**	**5.339**	5.787	5.500
Frat	42%	0.432	0.442	**0.414**	0.659	0.626	**0.622**	2.703	2.642	**2.608**
M5	37%	0.717	0.700	**0.662**	0.735	0.748	**0.702**	**0.866**	1.024	0.907
Favorita	42%	0.417	0.442	**0.395**	0.496	0.522	**0.472**	0.539	0.667	**0.525**

Table 8: RQ3, seen and unseen data. A fifth of the items, or of the stores, is kept out of training. WRMSSE on that fifth for the world model (WM), LightGBM (LGB) and their average (Avg); the cost of not training on them, which is that WRMSSE minus the WRMSSE on the same fifth of the same forecast trained on everything; and forecast over actual on the fifth. Means over windows. The last two columns name the forecast that is clearly lower on the kept-out fifth when a 90% bootstrap interval over windows of the difference excludes zero. Shading marks the lowest of the three: world model, LightGBM, average.

		WRMSSE, kept out			Cost of not training on them			Forecast over actual			Clearly lower	
Data set	Kept out	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg	WM or LGB	WM or Avg
Cheese	Items	**0.678**	0.731	0.687	0.011	0.058	0.035	0.99	0.93	0.96	World model	World model
Cheese	Stores	0.635	0.657	**0.612**	0.001	0.047	0.014	1.05	0.94	1.00	World model	Average
Snacks	Items	**0.626**	0.753	0.663	0.012	0.162	0.075	0.89	0.68	0.78	World model	World model
Snacks	Stores	**0.611**	0.700	0.621	0.001	0.102	0.034	1.00	0.72	0.86	World model	Neither
Frat	Items	**0.546**	0.843	0.654	0.008	0.311	0.135	0.95	0.66	0.81	World model	World model
Frat	Stores	**0.590**	0.779	0.634	0.007	0.205	0.071	0.99	0.77	0.88	World model	World model
M5	Items	**0.862**	0.918	0.871	0.040	0.083	0.053	1.04	0.84	0.94	World model	World model
M5	Stores	0.874	0.886	**0.869**	0.022	0.020	0.020	0.99	0.93	0.96	Neither	Neither
Favorita	Items	**0.709**	0.899	0.768	0.010	0.171	0.070	1.01	0.77	0.89	World model	World model
Favorita	Stores	**0.729**	0.766	0.731	0.020	0.017	0.020	0.97	0.98	0.97	World model	Neither

help.

Two parts of this rule were reached differently, and the difference matters to a reader who would use it. The average for series with history was the recommendation of the first revision and holds on the tuned models. The exception was tested as part of a wider rule written down before it ran, which also gave the world model every series with less than 26 weeks of history; that half added nothing on the data with history (it was level with the average on eight data sets and worse on snacks, 0.970 against 0.884), and only the new-item half held. The rule as stated is therefore drawn from these results.

Table 9: RQ4, transferability. WRMSSE overall / item by store of each tuned model trained on its own data set and trained on it together with other data sets (each Dominick's category with the other five and Frat; Frat with the six Dominick's categories; M5 with Favorita and Favorita with M5). The last row carries M5's tuned settings unchanged to Favorita, to be read against Favorita's own-data row.

	World model		LightGBM	
Data set	Own data	Pooled	Own data	Pooled
Analgesics	0.695 / 0.669	0.691 / 0.671	0.701 / 0.657	0.936 / 0.723
Bottled juice	0.573 / 0.615	0.613 / 0.610	0.576 / 0.626	0.582 / 0.653
Cheese	0.462 / 0.639	0.462 / 0.632	0.512 / 0.618	0.573 / 0.650
Frozen entrees	0.351 / 0.422	0.356 / 0.431	0.335 / 0.396	0.328 / 0.406
Soft drinks	0.744 / 0.600	0.885 / 0.747	0.721 / 0.570	0.805 / 0.594
Snacks	1.009 / 0.579	1.038 / 0.572	0.865 / 0.565	0.918 / 0.610
Frat	0.490 / 0.568	0.492 / 0.570	0.494 / 0.563	0.548 / 0.626
M5	0.709 / 0.838	0.733 / 0.840	0.697 / 0.858	0.851 / 0.876
Favorita	0.428 / 0.690	0.448 / 0.699	0.448 / 0.722	0.459 / 0.738
Favorita, M5's settings	0.447 / 0.692		0.450 / 0.723	

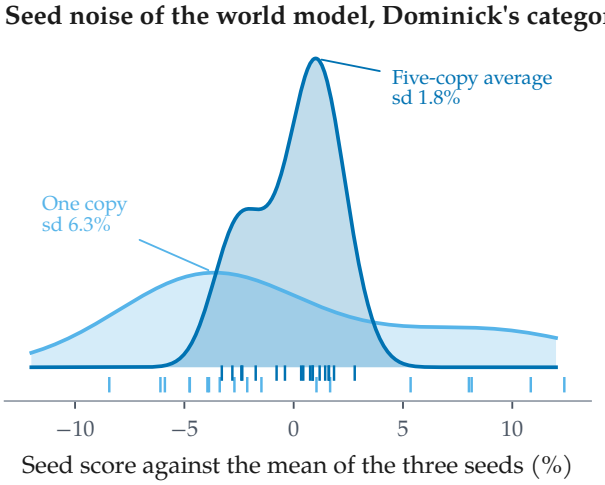


Figure 8: The starting settings' three seed scores on each Dominick's category, as a share of their mean, for a single world-model fit and for the five-copy average. Ticks are the scores.

The rule is a finding on these nine data sets, not a ranking of methods for any retailer. A retailer, or a researcher, adopting the framework should rerun the full set of comparisons on its own data, including the models and combinations that lost here: the pretrained models, the Temporal Fusion Transformer, the three-model ensemble, pooling across retailers and the forecast by history. Their order may change on data with other levers, other lines and other history; the code to do so is released with the paper. Section 12 sets out what the study does not cover, including forecasts longer than 8 weeks or 28 days.

9 Supporting analyses

The analyses below were run on the settings in use before the search of Section 7, and are kept for what they show about how the models use the plan; the numbers that answer the research questions are those of Section 8. All are means over the rolling windows; §9.10 gives the final windows of the first version.

9.1 Which model at which grain

Table 10 gives, for each grain at which a decision is made, the model with the lowest error and how many models the 90% confidence set kept with it. Every statement here rests on four to six windows per data set, and the windows are not independent: each forecast is made



Figure 9: Each model's starting and final settings on each data set. Open circles are the starting settings and filled circles the final settings, on the search windows (top) and the held-out window (bottom).

Table 10: On these data, which model had the lowest error depended on the grain, and the coarser the grain the more models the windows did not separate. For each decision grain and data set: the model with the lowest mean WRMSSE over the rolling windows, and in brackets how many of the nine models compared are in the 90% model confidence set (models four to six windows did not separate from the best; not a finding that they are equally good). Ens.: the equal-weight ensemble of the world model, LightGBM and TimesFM with the plan; World: world model; Lift: lift model; LGB: LightGBM; TFM+p: TimesFM with the plan. On Dominick's each data set is one category, so the category and the chain total are the same series. Favorita: the category is the product family. Daily on M5 and Favorita, weekly elsewhere.

Grain	Decision made there	*Cheese*	*Snack crackers*	*Soft drinks*	*Bottled juice*	*Analgesics*	*Frozen entrees*	*Frat*	*M5*	*Favorita*
Item by store	Store orders and inventory	LGB (1)	Ens. (1)	Ens. (2)	Ens. (1)	LGB (2)	Ens. (2)	Ens. (1)	Ens. (1)	Ens. (1)
Item, all stores	Supplier funding and deals	Ens. (3)	Ens. (2)	LGB (2)	Ens. (1)	Ens. (2)	Ens. (3)	Ens. (2)	Ens. (1)	Ens. (1)
Category by store	Store-level category performance	LGB (6)	LGB (9)	Ens. (3)	Ens. (4)	Ens. (2)	LGB (4)	Ens. (2)	LGB (4)	Ens. (2)
Category, all stores	Category sales and margin plans	Ens. (5)	Lift (9)	LGB (4)	Ens. (5)	Ens. (7)	LGB (4)	Ens. (4)	World (4)	Ens. (2)
Chain total	Financial plan	Ens. (5)	Lift (9)	LGB (4)	Ens. (5)	Ens. (7)	LGB (4)	TFM+p (4)	World (4)	Ens. (3)

from all earlier data, so adjacent windows share most of their training history, and errors that come from the same season or the same stretch of trading are correlated across windows. A model alone in the confidence set is a model that stood apart in these windows; a set of several is a set these windows did not separate, not a set of equally good models. Two patterns appear across the data sets.

At the fine grains one or two models stood apart. For an item in one store, the grain of store orders, the equal-weight average had the lowest error on seven of nine data sets and LightGBM on the other two (cheese and analgesics); the confidence set held a median of 1 model and never

more than 2. For an item across all stores, the grain of supplier funding, the average was lowest on eight and LightGBM on one, with a median of 2 models in the set. Neither the world model nor any pretrained model was in the set at these two grains except TimesFM with the plan and the world model once each for an item across stores.

At the coarse grains these windows did not separate the models. For a category by store, a category across stores and the chain total, the lowest-error model was the average on five, five and four data sets, LightGBM on four, two and two, and the world model, the lift model or TimesFM with the plan on the rest. A median of 4 of nine models were in the confidence set at each of these grains, and up to 9. A total is one series, so each window adds one error per period, and four to six windows leave too few to separate the plan-aware models there; more windows might. The world model was in the set at the chain total on eight data sets and the average on all nine.

Figure 10 compares the three core models at each grain against the average. LightGBM stays within a few percent of the average at every grain on almost every data set; cheese is the exception. TimesFM with the plan is close to the average for items and falls behind it as the grain coarsens. The world model is a few percent behind the average at every grain in the median, ahead of it at the M5 category and total, and far behind in cheese.

The decision costs agree (§9.9): the average or LightGBM had the lowest category-by-store planning cost on seven of eight data sets, whether the error is taken period by period or over the whole window, and the world model only on M5. The sections that follow look at each part of this picture: the fine grains and the average (§9.2, §9.3), the world model (§9.4), how the models use the plan (§9.5), the pretrained models (§9.6), the data set kept out of development (§9.7), intervals, decisions and reconciliation.

9.2 Item by store: the equal-weight average

The equal-weight ensemble has the lowest item-by-store error on seven of the nine data sets (Table 11, Figure 11a). In the other two, analgesics and cheese, LightGBM is lower: 0.647 against 0.650 in analgesics, where both are in the model confidence set, and 0.623 against 0.934 in cheese, the one data set where LightGBM is alone in the set. The ensemble is in the 90% model confidence set item by store on eight data sets and alone in it on five (snack crackers, bottled juice, Frat, M5 and Favorita). At the chain total it is in the set on all nine, and it has the lowest error on four; elsewhere the lowest belongs to LightGBM in frozen entrees; LightGBM in soft drinks; the lift model in snack crackers; TimesFM with the plan on Frat; the world model on M5.

By the paired test over windows, the ensemble is more accurate than LightGBM item by store on four of the eight data sets with paired results (bottled juice, snack crackers, Frat and M5; M5: 0.834 against 0.861) and at the total on one (analgesics). It is more accurate than the world model item by store on eight and at the total on two (cheese and Frat). It is less accurate than either at either level on none. On mean error, the Temporal Fusion Transformer, Chronos-Bolt and TimesFM from history alone are behind the ensemble item by store on every data set where they ran; they were not compared with it by a paired test.

The cheese result comes from one window, which we looked for after reading the result. In one of the six cheese windows the world model's item-by-store WRMSSE was 7.12, against a median of 0.63 over the six, and the lift model's 3.57; the ensemble, a third of whose forecast is the world model's, inherits part of that error.

9.3 What each model adds to the average

Each member earns its place on some data sets and none on all (Table 12). Without the world model the ensemble is less accurate item by store on four (bottled juice, frozen entrees, M5 and Favorita) and at the total on three (bottled juice, frozen entrees and M5); on Frat at the total it is more accurate without it (-0.040). Without LightGBM it is less accurate item by store on eight and at the total on four. Without TimesFM with the plan it is less accurate item by store on five, at the total on one, and more accurate at the total on two (M5 and Favorita). LightGBM is the member the ensemble can least do without.

Replacing the world model with the lift model, the same network without the carried state, tests what the carried state adds. On bottled juice, frozen entrees and M5, where the world model adds item by store, and at the total on bottled juice, frozen entrees and M5, we did not detect a difference between the ensemble with the lift model and with the world model. That is not evidence that the two are equal, and the lift model is one replacement, so it does not show that any third model would do as well. It is less accurate with the lift model item by store on three data sets (analgesics, soft drinks and Favorita) and at the total on two (analgesics and Favorita), but by the rule written down for study A the world model adds to the ensemble only where the ensemble beats the pair without it, and on soft drinks and analgesics it does not. So we detected the carried state adding to an ensemble that gains from the world model only on Favorita item by store, and at the total on none.

On Frat the two networks did not differ only in the car-

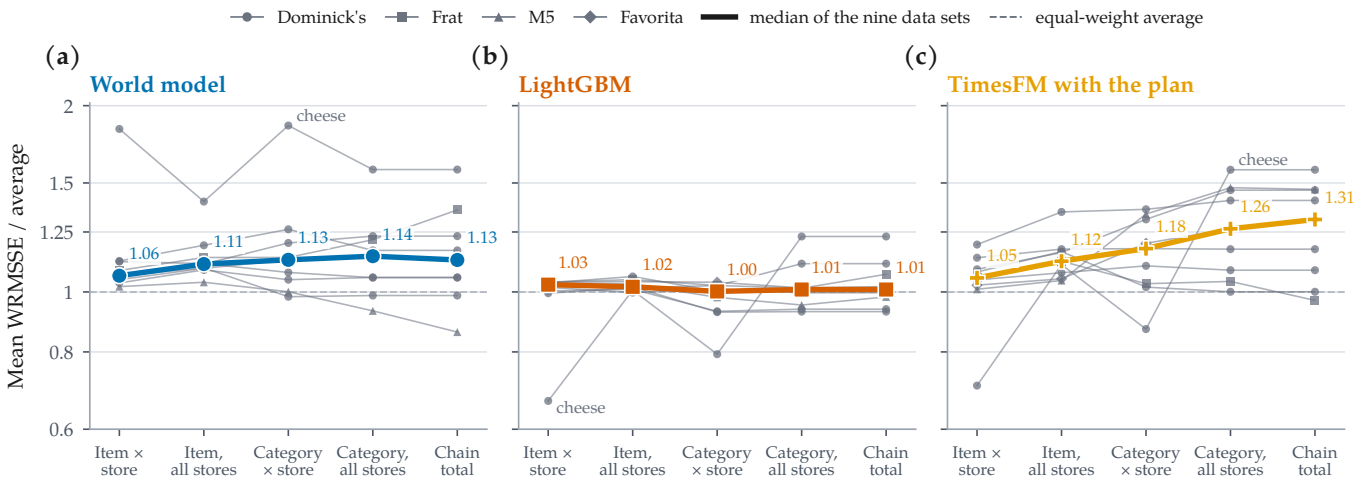


Figure 10: Against the average, LightGBM was close at every grain, TimesFM with the plan fell behind as the grain coarsened, and the world model was a few percent behind with a wide spread. Mean WRMSSE over the rolling windows of (**a**) the world model, (**b**) LightGBM and (**c**) TimesFM with the plan, divided by the equal-weight average's, at the five decision grains of Table 10. Thin grey lines: one per data set (marker: data source); thick line and labels: the median over the nine data sets; dashed: the average. Log scale. On Dominick's the category and the chain total are the same series.

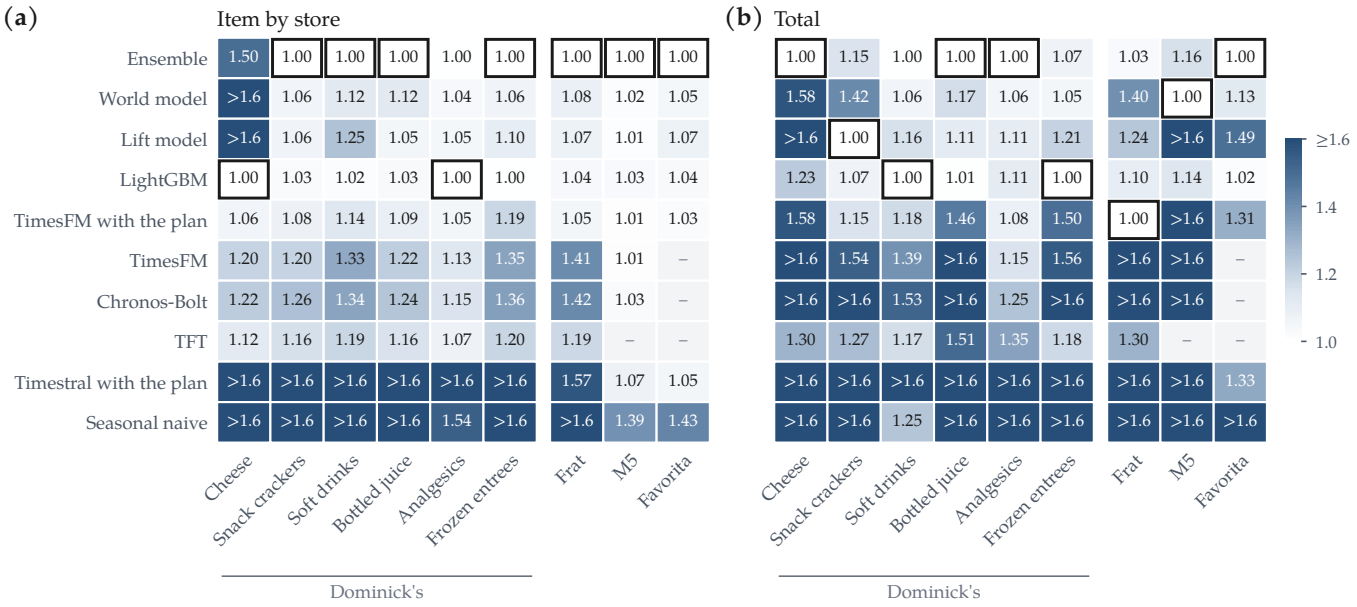


Figure 11: Item by store the average or LightGBM was best on every data set; at the total the best model changed from data set to data set. Mean WRMSSE over the rolling windows of every model divided by the best in its column (1.00: the best, outlined), (**a**) item by store and (**b**) at the chain total. Dash: not run. Six windows on each Dominick's category, four on Frat, M5 and Favorita.

ried state: the lift model's settings, carried over from the first version, add readout dropout and a larger weight decay that the world model does not have. We reran it with the world model's settings (study G), so that only the transition differs. On its own it is then less accurate than the world model (item by store 0.602 against 0.595; total 0.473 against 0.457), where the regularised lift model had a lower mean total error (0.405), a difference the paired test did not detect. In the ensemble in the world model's place we again detect no difference (item by store 0.003, interval −0.001 to 0.008; total 0.007, interval −0.020 to 0.038). The Frat rows of Table 12 keep the regularised lift model, as run.

Equal weights are hard to beat. Weights chosen on the earlier windows of each data set, for item by store, for the total or for all levels, and scored on the next window, are more accurate than equal weights in five of 54 comparisons and less accurate in 13. The fitted weights do say where each member is useful: chosen for the total, the world model's mean weight is 0.64 in analgesics and 0.62 on M5, against 0.09 and 0.32 chosen for item by store.

Table 11: Over the rolling windows the equal-weight ensemble has the lowest item-by-store error on seven of nine data sets, is in the 90% model confidence set item by store on eight and at the total on nine. Mean WRMSSE over the rolling windows (lower is better); W: windows. Bold: the lowest in the row. °: in the 90% model confidence set of the row (Hansen, Lunde and Nason, range statistic, bootstrap over windows; Timestral, added in study F, is not in the comparison). Ensemble: the equal-weight mean of the world model, LightGBM and TimesFM with the plan, fixed before any rolling result. Trained models are one fit each (§6). The Temporal Fusion Transformer was not run on M5 or Favorita, Chronos-Bolt, TimesFM and TFT not on Favorita. Timestral+plan is Timestral-28M with the plan (study F).

Data	W	*Ensemble*	*World model*	*Lift model*	*LightGBM*	*TimesFM+plan*	*TimesFM*	*Chronos-Bolt*	*TFT*	*Timestral+plan*	*Seas. naive*
Item by store											
Cheese	6	0.934	1.713	1.119	**0.623**°	0.659	0.749	0.762	0.696	1.956	1.117
Snack crackers	6	**0.562**°	0.595	0.598	0.578	0.605	0.677	0.708	0.654	1.944	1.010
Soft drinks	6	**0.536**°	0.600	0.672	0.545°	0.609	0.711	0.720	0.640	0.872	0.960
Bottled juice	6	**0.585**°	0.656	0.616	0.605	0.636	0.712	0.728	0.676	1.459	1.079
Analgesics	6	0.650°	0.672	0.677	**0.647**°	0.679	0.734	0.745	0.693	2.111	0.997
Frozen entrees	6	**0.408**°	0.433	0.449	0.410°	0.487	0.549	0.554	0.488	1.048	0.710
Frat	4	**0.549**°	0.595	0.590	0.570	0.578	0.775	0.778	0.652	0.864	1.077
M5	4	**0.834**°	0.851	0.845	0.861	0.842	0.842	0.858	–	0.894	1.163
Favorita	4	**0.694**°	0.726	0.744	0.719	0.712	–	–	–	0.731	0.992
Total											
Cheese	6	**0.371**°	0.585°	0.617°	0.456°	0.584	0.718	0.790	0.481°	4.606	1.175
Snack crackers	6	1.181°	1.454°	**1.026**°	1.097°	1.181°	1.581°	4.410°	1.301°	16.272	1.790°
Soft drinks	6	0.834°	0.879°	0.965°	**0.831**°	0.977	1.157	1.270	0.971	4.181	1.035
Bottled juice	6	**0.525**°	0.613°	0.585°	0.530°	0.766°	0.898	0.897	0.793	6.257	1.390
Analgesics	6	**0.782**°	0.826°	0.868°	0.869°	0.848°	0.900°	0.979	1.057°	31.256	1.796
Frozen entrees	6	0.342°	0.338°	0.387°	**0.321**°	0.481	0.500	0.578	0.380	2.158	0.620
Frat	4	0.337°	0.457	0.405°	0.360°	**0.327**°	0.541	0.605	0.425	1.048	0.964
M5	4	0.706°	**0.609**°	1.163°	0.693°	1.035	1.000	1.333	–	1.025	1.087
Favorita	4	**0.303**°	0.341°	0.452	0.309°	0.397	–	–	–	0.403	0.501

Figure 12c shows the item-by-store error is flat across a wide range of the world model's weight on most data sets, which is why fitting the weights gains little.

9.4 The three core models, window by window

Figure 13 shows every rolling window. Item by store, the average had a lower error than LightGBM in 35 of 48 windows (median ratio 0.97), TimesFM with the plan in 12 (median 1.03) and the world model in 7 (median 1.04); the world model's worst window, in cheese, was 9.6 times LightGBM's error. At the total the world model was more accurate than LightGBM in 22 of 48 windows and TimesFM with the plan in 14: window by window neither is consistently behind LightGBM there, although their means are (Figure 13b, c).

On its own the world model is behind LightGBM item by store, and at the total we did not detect a difference on most data sets. By the paired test LightGBM is more accurate item by store on six of eight data sets (analgesics, bottled juice, cheese, soft drinks, snack crackers and Frat) and at the total on one (Frat); the world model is more accurate than LightGBM on none at either level. Its lowest error is at the M5 chain total, 0.609 against 0.693 for LightGBM and 0.706 for the ensemble, but the windows differ enough that the difference is not outside its interval (LightGBM minus the world model 0.085, interval −0.247 to 0.343).

Against the lift model (on Frat the regularised one, §9.3), the world model is more accurate item by store on one (soft drinks), less accurate on two (bottled juice and M5), and no difference is detected at the total on any of the eight. Taken alone, we did not detect an effect of carrying the state forward on accuracy in most data sets. What it may add shows in what it adds to the ensemble (§9.3) and at the M5 totals, where it has the lowest total error and the lowest category-plan cost over the window (§9.9).

Table 12: Removing the world model makes the ensemble less accurate item by store on four data sets and at the total on three; replacing it with the lift model, the same network without the carried state, does so on three and two. Mean WRMSSE of each variant minus the ensemble's over the rolling windows (study A, added after the rolling results); above zero the ensemble is more accurate. Bold: the ensemble is more accurate and the 95% interval from the bootstrap over windows excludes zero; underlined: the variant is, with the same test. No X: the equal-weight mean of the other two members (−World: without the world model; −LGB: without LightGBM; −TFM+p: without TimesFM with the plan). Lift: the world model replaced by the lift model. Fitted: weights chosen on the earlier windows of the data set (lowest mean WRMSSE over all levels), scored on the next, so one window fewer.

	Item by store					Total				
Data	−World	−LGB	−TFM+p	Lift	Fitted	−World	−LGB	−TFM+p	Lift	Fitted
Cheese	−0.328	**0.199**	**0.190**	−0.196	**0.015**	0.072	**0.056**	0.029	0.021	0.056
Snack crackers	0.002	**0.018**	0.004	0.002	0.004	−0.095	0.118	0.013	−0.145	−0.073
Soft drinks	0.008	**0.023**	0.009	**0.015**	−0.006	0.025	**0.045**	−0.022	−0.007	−0.008
Bottled juice	**0.012**	**0.016**	**0.015**	0.002	0.000	**0.103**	0.031	−0.042	0.051	−0.015
Analgesics	0.000	**0.014**	−0.001	**0.003**	**0.008**	0.019	−0.006	0.038	**0.025**	0.012
Frozen entrees	**0.012**	**0.019**	−0.002	0.005	−0.001	**0.025**	**0.034**	−0.026	0.013	−0.015
Frat	0.002	**0.010**	**0.012**	0.001	**0.007**	−0.040	**0.021**	**0.051**	−0.017	−0.022
M5	**0.005**	0.003	**0.006**	−0.001	**0.013**	**0.113**	0.050	−0.112	0.000	−0.037
Favorita	**0.007**	**0.006**	**0.007**	**0.010**	**0.007**	0.037	0.013	−0.015	**0.067**	0.004

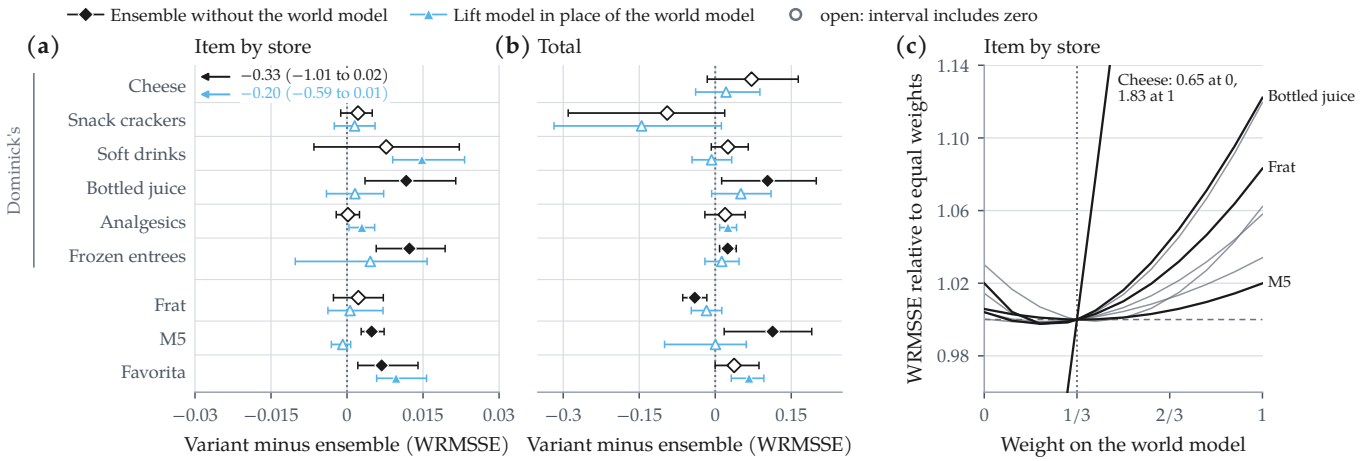


Figure 12: Taking the world model out of the ensemble costs accuracy on some data sets; replacing it with the same network without the carried state costs less often. (**a, b**) Mean WRMSSE of the ensemble without the world model, and with the lift model in its place (on Frat the lift model also had readout dropout and weight decay; §9.3), minus the ensemble's, with 95% intervals from the bootstrap over windows; right of zero the ensemble with the world model is more accurate; filled markers: the interval excludes zero. (**a**) Item by store. (**b**) Total. (**c**) Item by store: the ensemble's WRMSSE as the world model's weight goes from 0 to 1 (the other two sharing the rest equally), relative to equal weights (dotted line at one third), one line per data set. Study A, saved forecasts only.

9.5 How the models use the plan

All three trained models use the plan (Table 13). With the whole plan set to "nothing planned" the total error on Dominick's rises by 116% for the world model, 97% for the lift model and 149% for LightGBM, and the forecast in planned periods changes by −59%, −58% and −58%. Price carries more of it than the promotion flag on Dominick's and Frat. On M5 prices rarely change, and the plan inputs that matter are event and SNAP days: occluding SNAP days raises the world model's total error by 58% and LightGBM's by 16%, and occluding event days by 16% against 0%. The world model reads these calendar inputs more than LightGBM does, which fits its lower error at the M5 total. On Frat the promotion flag adds little to the world model once price is given: occluding it changes the world model's error by −1% item by store and 20% at the total, against 99% for the lift model. Permuting an input within each series across the window, rather than removing it, costs more item by store and less at the total: permuting price raises the Dominick's item-by-store error by 24% for the world model

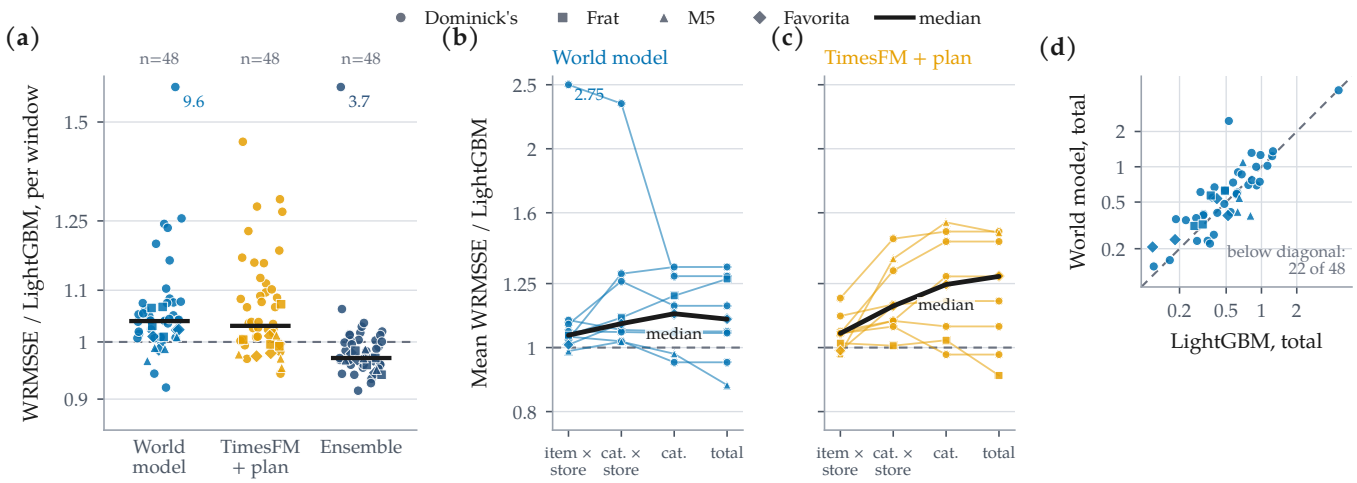


Figure 13: Window by window, the average was more accurate than LightGBM item by store in most windows, and the world model and TimesFM with the plan in few; both fell further behind LightGBM as the grain coarsened. (**a**) Item-by-store WRMSSE of the world model, TimesFM with the plan and the average, each divided by LightGBM's in the same window, one point per rolling window and data set ($n = 48$; marker: data source; bar: median; log scale; values above the axis written). (**b, c**) Mean WRMSSE over LightGBM's at four grains, one line per data set and the median (thick) for the world model and TimesFM with the plan. (**d**) Total WRMSSE of the world model against LightGBM, one point per window, log scales; below the diagonal the world model is more accurate.

Table 13: All three trained models use the plan. On the weekly data price carries most of it; on M5 the world model relies on event and SNAP days more than LightGBM does. Input occlusion on the saved models (study B): one plan input is set to "nothing planned" (regular price, lever off) in the forecast periods and the forecast recomputed, with no refitting. Change in WRMSSE item by store and at the total, in percent (above zero: the forecast gets worse without the input), and the change in forecast units, in percent, in the periods where the input was on. Mean over the rolling windows (LightGBM on M5: the last two).

	Item by store (%)			Total (%)			Units where on (%)		
Input occluded	World	Lift	LGB	World	Lift	LGB	World	Lift	LGB
Dominick's									
Whole plan	15	13	26	116	97	149	−59	−58	−58
Price	8	6	20	75	59	115	−45	−41	−49
Promotion	4	2	3	41	38	34	−34	−40	−26
Frat									
Whole plan	36	36	40	229	268	262	−49	−47	−46
Price	17	16	18	126	152	138	−38	−36	−34
Promotion	−1	6	4	20	99	34	−16	−30	−18
Feature	1	2	4	7	23	16	−28	−28	−29
Display	0	0	2	4	9	4	−18	−19	−17
Price reduction only	1	1	1	9	10	3	14	16	10
M5									
Whole plan	1	1	1	45	14	15	−8	−19	−4
Price	0	0	0	3	−26	−1	−35	−60	18
Event days	0	1	0	16	18	0	15	20	2
SNAP days	1	2	0	58	30	16	−14	−14	−6

and 32% for LightGBM (Table 13 shows occlusion only).

Three tests on the saved world models show what the carried state does with the plan. *Timing*: shuffling the plan within each series, with its total exposure unchanged, raises item-by-store error in cheese by 29% for the world model and 26% for the lift model, and in snack crackers by 26% and 27%: both use when the plan falls, and the carried state adds little to that. *Persistence* (Figure 14a): after a promotion and price cut added in the first period, the world model's forecast stays higher in later periods, by 15% one period later in snack crack-

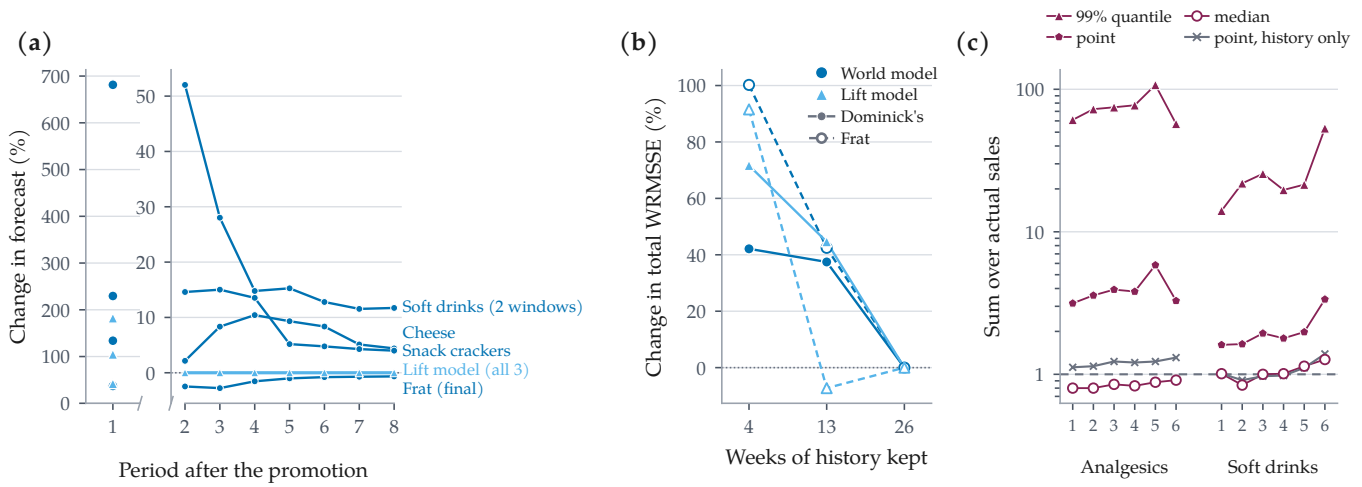


Figure 14: The world model carries a promotion forward as further lift and uses all 26 weeks of history it reads; Timestral given the plan puts extreme upper quantiles on a few series. **(a)** Persistence: the change in the forecast, in percent, at each later period after one promotion and a 20% price cut are added in the first period of the window, for the world model and the lift model (which has no carried state, so its later change is zero by construction). **(b)** History occlusion: the change in total WRMSSE, in percent, when only the last 4 or 13 of the 26 weeks of history the models read are kept (26: nothing removed). **(c)** Timestral with the plan on two Dominick's categories: the sum over all series of its median, its point forecast (the mean of its quantiles) and its 99% quantile, over actual sales, per rolling window; log scale, dashed line at 1. Study B and the mechanism tests use the saved models with no refitting; (c) from study F.

ers and 8% two periods later in cheese, and above zero through the window. In categories where shoppers stock up, sales after a promotion dip; the world model carries the promotion forward as further lift. On the Frat final window, the one Frat window tested, the effect is a small dip (−2% one period later). *State*: a ridge read-out from the rolled state predicts sales in the periods after the window no better than the same read-out from the state before it (R^2 0.92 against 0.91 in cheese, 0.64 against 0.64 on M5).

On the weekly data the world model and the lift model read the last 26 weeks of history, and removing the older part of it costs accuracy (Figure 14b): keeping only the last 13 raises the Dominick's total error by 37% for the world model and 45% for the lift model, and keeping only the last 4 by 42% and 72%. On M5, keeping only the last 28 days raises the world model's total error by 615% and the lift model's by 255%.

9.6 Pretrained models

On these data TimesFM with the plan was the most accurate model not trained on the data, and the plan matters to it: item by store it is ahead of TimesFM from history alone on every data set where both ran except M5, where their means agree to three decimals (Table 11), and most on the weekly promotion-driven data (soft drinks 0.609 against 0.711). LightGBM is more accurate than it item by store on six data sets and at the total on four, and TimesFM with the plan is more accurate than LightGBM on none. Chronos-Bolt, from history alone, is behind TimesFM from history alone on every data set item by store, and its total error in snack crackers is 4.410.

Timestral (study F, Table 14) is behind the ensemble on every data set, at both levels. Given the plan it is less accurate than TimesFM given the plan by the paired test over windows in 16 of 18 comparisons (every data set item by store and at the total), and not separated from it at the total on Favorita and the total on M5 (1.025 against 1.035 on M5); it is more accurate in none. From history alone it is less accurate than TimesFM from history alone in 10 of 16 comparisons and more accurate in none. This test was asked for by the plan and computed from the saved per-window scores after the first version of this section, by the bootstrap over windows only. Putting it in place of TimesFM makes the ensemble less accurate on Dominick's and Frat, not separated at the Favorita total and slightly less accurate item by store, and more accurate only at the M5 total (0.609 against 0.706); as written down before it ran, the paper's ensemble is unchanged and that swap is reported as a robustness result. The larger Timestral-209M, run on Frat to check the model size, is less accurate than the 28M model (1.036 against 0.864 item by store).

On Dominick's Timestral with the plan forecasts more than total sales, in analgesics several times more (Figure 14c). We checked the cause after reading the result. It is not a scale or input error: summed over all series, its median is 0.80 to 0.91 times actual sales in analgesics, and its history-only point forecast 1.12 to 1.31. Given

Table 14: Timestral, released two days after this revision began, is behind the ensemble on every data set, and putting it in place of TimesFM helps only at the M5 total. Mean WRMSSE over the rolling windows (Dominick's: mean of the six categories). Timestral-28M, zero-shot, the mean of its 21 quantiles as the point forecast; with the plan as known future inputs (TS+p) and from history alone (TS); TFM+p: TimesFM with the plan. Ens.: the ensemble; Ens. T: Timestral with the plan in place of TimesFM with the plan; Ens. 4: all four together. Study F, written down before it ran.

	Item by store						Total					
Data	Ens.	Ens. T	Ens. 4	TFM+p	TS+p	TS	Ens.	Ens. T	Ens. 4	TFM+p	TS+p	TS
Dominick's (six)	**0.612**	0.916	0.821	0.613	1.565	0.728	**0.672**	3.729	2.858	0.807	10.788	2.121
Frat	**0.549**	0.623	0.595	0.578	0.864	0.792	0.337	0.501	0.407	**0.327**	1.048	0.484
M5	**0.834**	0.846	0.842	0.842	0.894	0.861	0.706	**0.609**	0.699	1.035	1.025	1.014
Favorita	**0.694**	0.697	**0.694**	0.712	0.731	0.771	0.303	**0.298**	0.313	0.397	0.403	0.430

the plan, its 99% quantile sums to 57 to 107 times actual sales, and because the point forecast is the mean of its quantiles, a few series carry the error: the top 1% of series by excess hold 75% to 88% of it. The same happens more mildly in soft drinks (99% quantile 14 to 53 times sales) and Frat (3.4 to 6.9), and not on the daily data. The forecast rule was not changed after the result. Timestral's median sums much closer to sales here, but the point forecast was fixed in advance to the mean, as for Chronos-Bolt, because a median forecasts well below totals for items that often sell nothing; how the median would score was not tested.

9.7 A data set kept out of development

On Favorita, where nothing was tuned and every setting came from the M5 configuration, the ensemble has the lowest error at every level and is alone in the model confidence set item by store (0.694 against 0.719 for LightGBM and 0.726 for the world model); at the total LightGBM (0.309) and the world model (0.341) are in the set with it (0.303). The world model is behind LightGBM at every level, as on most of the other data sets, and still adds to the ensemble item by store: without it the ensemble's error rises by 0.007, and with the lift model in its place by 0.010. The lift model is far behind at the total (0.452). Favorita has no prices, so its weights are unit weights and its plan is promotions, holidays and paydays.

9.8 Intervals and ranking

The first conformal intervals covered far less than planned at the totals (95% intervals holding 0.27 of actual sales at the M5 total for the ensemble and 0.08 for the world model on Frat), because a level with one node has one earlier error per horizon per window, so the 2.5% and 97.5% quantiles were the smallest and largest of one to four numbers. With horizons pooled where errors are few (Table 15, Figure 15b) the ensemble's 95% interval holds 0.94 at the M5 total and 0.96 on Frat. Item by store the 95% intervals hold 0.93 to 0.98 of sales for all three models, and above 0.95 on M5 and Frat because rounding a narrow interval outwards to a whole unit widens it; the pinball loss, unchanged by rounding, is the measure to compare there, and it separates the three models by at most 0.03 on every data set. Scored on their own quantiles item by store, TFT's have a lower pinball loss than the ensemble's conformal intervals on seven of the seven data sets where it ran (0.16 against 0.18 in frozen entrees, 0.19 against 0.20 on Frat), and Chronos-Bolt's are behind them (0.42 on M5 against 0.32). Native quantiles are scored over every window and the conformal ones over all but the first, so the comparison is close but not exact.

The ensemble also ranks item-by-store changes better than any single model (Figure 15c). Across item-store series, the Spearman correlation between forecast and actual change is 0.61 on Dominick's, 0.46 on Favorita, 0.83 on Frat and 0.42 on M5, above every single model in the check on each; of the tenth of series it forecasts to grow most, 0.57, 0.51, 0.68 and 0.53 are in the actual top tenth, against 0.10 by chance. Over promoted periods LightGBM and the ensemble rank promotions best (0.67 and 0.66 on Dominick's). The ensemble's total forecast is within 6% of actual sales on every data source (0.94 on M5 is the furthest).

9.9 Decisions and reconciliation

Both decisions point to the ensemble or LightGBM (Table 16, Figure 15a). For the category plan the ensemble has the lowest cost on four of eight data sets (analgesics, bottled juice, soft drinks and Frat) and LightGBM on three (cheese, frozen entrees and snack crackers); on seven of the eight the lowest of them is ahead of the world model by a paired interval that excludes zero (Table 17). The world model is lowest only on M5, where by period it is not separated from the ensemble (-0.002, interval -0.006 to 0.003) and over the whole window it is clearly lowest (0.038 against 0.058 for the ensemble

Table 15: Rolling conformal intervals cover close to their level item by store, from 0.72 to 1.00 at the 95% level at the total, and their pinball losses item by store are within 0.03 of each other. Scaled pinball loss (SPL, lower is better) and the share of actual sales inside the 80% and 95% intervals, mean over the rolling windows after the first. Intervals from each model's own scaled errors in the earlier windows of the data set, pooled over horizons where a level has fewer than 50 earlier errors per horizon, ends rounded outwards to whole units except on Favorita (interval check of 16 October).

	Item by store									Total								
	Ensemble			World			LightGBM			Ensemble			World			LightGBM		
Data	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95	SPL	80	95
Cheese	0.22	0.75	0.94	0.25	0.74	0.93	0.23	0.75	0.94	0.12	0.68	0.88	0.20	0.72	0.97	0.16	0.93	0.95
Snack crackers	0.23	0.74	0.94	0.24	0.73	0.94	0.23	0.73	0.93	0.53	0.60	0.72	0.80	0.57	0.60	0.42	0.75	0.88
Soft drinks	0.23	0.75	0.94	0.25	0.74	0.94	0.23	0.75	0.93	0.27	0.68	0.80	0.28	0.80	0.82	0.26	0.65	0.85
Bottled juice	0.23	0.73	0.93	0.26	0.72	0.93	0.24	0.73	0.93	0.15	0.82	0.93	0.20	0.80	0.95	0.15	0.93	0.97
Analgesics	0.25	0.91	0.98	0.25	0.91	0.98	0.24	0.91	0.98	0.25	0.85	0.88	0.29	0.85	0.95	0.27	0.88	0.95
Frozen entrees	0.18	0.69	0.93	0.18	0.69	0.93	0.17	0.70	0.93	0.12	0.90	0.97	0.12	0.88	1.00	0.10	0.82	0.90
Frat	0.20	0.87	0.97	0.22	0.87	0.97	0.21	0.87	0.97	0.11	0.75	0.96	0.15	0.83	0.88	0.14	0.62	0.83
M5	0.32	0.94	0.98	0.33	0.94	0.98	0.33	0.94	0.98	0.18	0.71	0.94	0.19	0.76	0.86	0.14	0.80	0.98
Favorita	0.24	0.79	0.95	0.25	0.79	0.95	0.24	0.79	0.95	0.07	0.83	0.98	0.10	0.70	0.93	0.07	0.86	1.00

Table 16: For category plans and for store orders the ensemble or LightGBM has the lowest cost on every data set except the M5 category plan. Cost relative to seasonal naive (below 1 is better), mean over the rolling windows. Category plan: absolute error of each category-by-store total in each period, in dollars, over sales in dollars (corrected in study G; Table 17). Store orders: order the model's 80% interval quantile for each item, store and period; each unit over costs 1 and each unit short costs 4. Simulations under stated costs, not business results. Bold: the lowest in the group.

	Category plan					Store orders (ratio 0.8)				
Data	Ens.	World	Lift	LGB	TFM+p	Ens.	World	Lift	LGB	TFM+p
Cheese	0.53	0.94	0.69	**0.45**	0.50	0.52	0.55	0.57	**0.52**	0.62
Snack crackers	0.60	0.78	0.76	**0.52**	0.62	0.72	0.77	0.77	**0.71**	0.76
Soft drinks	**0.72**	0.78	0.93	0.74	0.87	0.59	0.61	0.64	**0.57**	0.69
Bottled juice	**0.38**	0.48	0.46	0.40	0.45	**0.48**	0.49	0.51	0.49	0.54
Analgesics	**0.46**	0.48	0.49	0.48	0.51	0.68	0.70	0.71	**0.68**	0.71
Frozen entrees	0.58	0.56	0.67	**0.50**	0.83	0.51	0.52	0.60	**0.51**	0.63
Frat	**0.50**	0.57	0.56	0.51	0.52	**0.50**	0.53	0.52	0.52	0.56
M5	0.64	**0.63**	1.29	0.65	0.84	**0.82**	0.83	0.88	0.85	0.83

and 0.074 for LightGBM), in line with its lowest total error there (§9.4). Under every rule written down for the sensitivity check (the first half of the window, the error of the whole window's total, units in place of dollars, every node weighted equally) the lowest cost on the seven other data sets belongs to the ensemble, LightGBM or, once, TimesFM with the plan, and the world model is lowest on M5 at most. The node bootstrap covers the choice of stores and categories within a window, not the variation between windows, which with four to six windows remains the larger uncertainty; by period the world model costs less than the lowest other model in no more than two windows on any data set.

A correction. The first version of this cost divided each node's error by its sales in the window, with a floor of 10^{-9}, and put the world model first on six of eight data sets. On Dominick's 4 or 5 of the 558 category-by-store nodes and windows had no sales (a store closed or not reporting), and those nodes entered at about 10^{9} times their error, so the result measured which model forecast closest to zero for a store that had stopped selling. Frat and M5 have no such nodes; their costs move slightly with the change to dollars and their ranking is unchanged. The plan defines the cost as the absolute error of category-by-store totals weighted by dollar sales; it is now computed as that error in dollars over sales in dollars, with each node's units valued at its average price before the window (study G).

For store orders at a critical ratio of 0.8, LightGBM has the lowest cost on five (analgesics, cheese, frozen entrees, soft drinks and snack crackers) and the ensemble on three (bottled juice, Frat and M5); the world model

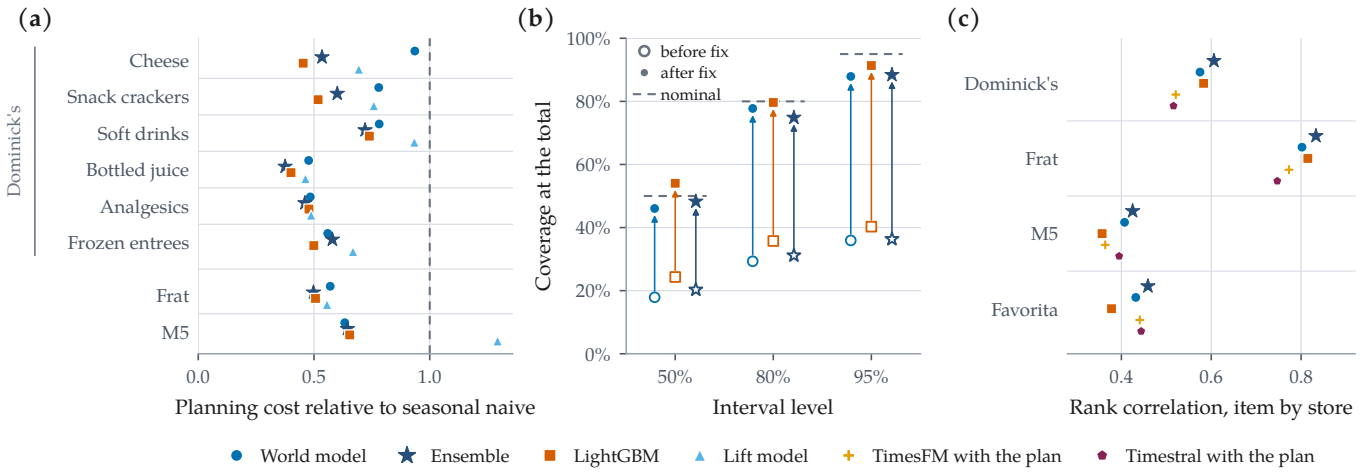


Figure 15: The ensemble or LightGBM has the lowest category-plan cost on seven of eight data sets, pooling fixed the interval coverage at the totals, and the ensemble ranks changes best. **(a)** Category-plan cost (error of category-by-store totals in dollars, corrected in study G) relative to seasonal naive (dashed line at 1), mean over the rolling windows. **(b)** Share of actual sales inside the 50%, 80% and 95% intervals at the chain total, mean over the eight data sets scored both before the interval fix (open) and after it (filled); dashes: the nominal level. **(c)** Spearman correlation across item-store series between forecast and actual change, mean over the rolling windows (Dominick's: over its six categories).

Table 17: With the cost corrected, the world model had the lowest category-plan cost only on M5; the ensemble or LightGBM was lowest elsewhere under every rule but one. On M5 the world model is not separated from the ensemble by period and is lower in all four windows over the whole window; the exception elsewhere is cheese over the first half, where TimesFM with the plan is lowest. Category-by-store planning cost, mean over the rolling windows. Lowest other: the model other than the world model with the lowest cost by period, in dollars. World minus it: the world model's cost minus that model's, with a 95% interval from 5,000 resamples of category-by-store nodes within each window, the same nodes for both models (positive: the world model costs more); this interval does not cover variation between windows, for which the next column counts the windows in which the world model cost less. The last five columns give the lowest-cost model of the five under each rule: absolute error of each period's total in dollars (the cost used in Table 16); the same over the first half of the window; error of the total over the whole window; each unit of error counted alike; and the mean over nodes of each node's error over its sales. Study G.

				Lowest cost under each rule				
Data	**Lowest other**	**World minus it**	**World lower**	By period	First half	Window total	Units	Equal nodes
Cheese	LGB	0.129 (0.046 to 0.212)	2/6	LGB	TFM+p	LGB	LGB	Ens.
Snack crackers	LGB	0.100 (0.081 to 0.122)	2/6	LGB	LGB	LGB	LGB	LGB
Soft drinks	Ens.	0.017 (0.014 to 0.019)	1/6	Ens.	Ens.	Ens.	Ens.	Ens.
Bottled juice	Ens.	0.033 (0.029 to 0.037)	0/6	Ens.	Ens.	LGB	Ens.	Ens.
Analgesics	Ens.	0.005 (0.003 to 0.007)	1/6	Ens.	Ens.	LGB	Ens.	Ens.
Frozen entrees	LGB	0.024 (0.020 to 0.028)	2/6	LGB	LGB	LGB	LGB	LGB
Frat	Ens.	0.021 (0.019 to 0.023)	0/4	Ens.	Ens.	Ens.	Ens.	Ens.
M5	Ens.	−0.002 (−0.006 to 0.003)	2/4	World	World	World	World	LGB

on none. At 0.95 the world model is lowest on bottled juice and M5, by less than 0.02.

For promotion decisions, in the companion paper's generated worlds where the true lifts are known, deciding at the median promotion cost with the world model's predicted lift loses 1.7% to 2.1% of the value of deciding with the true lifts across three seeds, and with LightGBM's 1.8%; in the world where merchants promote into strong weeks, 2.0% to 2.5% against 1.6%. A world model that cannot see the promotion levers loses 77% and 98%.

Reconciliation does not replace a model trained on the plan (Table 18). The best reconciled ETS forecast on Frat is top-down, at 0.806 item by store against 0.570 for LightGBM. Combining a trained model's item forecasts with ETS above them by weighted least squares lowers the Frat total error (LightGBM 0.304 against 0.360 unreconciled; world model 0.343 against 0.457) and raises the M5 total error (world model 0.754 against 0.609). On Dominick's it raises LightGBM's total error in six of six categories (cheese 1.526 against 0.456), so it is not a fix for the weekly totals. There the best reconciled ETS fore-

Table 18: Reconciled ETS forecasts are far behind the trained models; reconciling a trained model's item forecasts with ETS above them helps at the Frat total and hurts on Dominick's and M5. Mean WRMSSE over the rolling windows (Dominick's: mean of the six categories). MinT with a shrunk covariance was computed only where the hierarchy has at most 6,000 nodes (Frat), after the fix for nodes with no in-sample error; its first run gave a total WRMSSE of 247.170 (§9.9).

	Dominick's		Frat		M5	
Method	I×S	Total	I×S	Total	I×S	Total
ETS, bottom-up	0.736	1.165	0.909	0.566	0.898	1.271
ETS, top-down	0.753	0.924	0.806	0.487	0.917	0.902
ETS, OLS	0.778	1.177	0.953	0.520	0.888	0.902
ETS, WLS structural	0.748	1.125	0.912	0.538	0.885	0.908
ETS, WLS variance	0.853	1.750	0.903	0.511	0.884	0.946
ETS, MinT shrink	–	–	0.926	0.610	–	–
LightGBM + ETS, WLS variance	0.720	1.483	0.581	0.304	0.855	0.763
World model + ETS, WLS variance	0.786	1.666	0.602	0.343	0.849	0.754
LightGBM, unreconciled	0.568	0.684	0.570	0.360	0.861	0.693
World model, unreconciled	0.778	0.782	0.595	0.457	0.851	0.609
Ensemble, unreconciled	0.612	0.672	0.549	0.337	0.834	0.706

cast at the total, whichever method gives it, is behind LightGBM and the ensemble in every category. Item by store, ETS bottom-up is behind LightGBM in every category, and ahead of the world model and the ensemble only in cheese, where one window inflates their means (0.791 against 1.713 and 0.934). MinT with a shrunk covariance first gave a total WRMSSE of 247.170 on Frat. The cause was numerical: 47 of the 4,349 Frat nodes had no in-sample error (constant histories), so the covariance was singular, and the solve gave item forecasts of the order of 10^7 of both signs, which the rule that forecasts are not below zero then turned into very large totals. With those nodes given the smallest variance of the others and the shrinkage intensity computed rather than fixed, MinT-shrink gives 0.610 at the Frat total and 0.926 item by store; at the total it is behind every other ETS method and behind LightGBM (0.360); the other methods are unchanged.

9.10 The final windows of the first version

The first version scored each data set once, on its final window; those results are in Supplement C. They agree with the rolling windows where the two overlap: on the final windows the ensemble is lowest or within 0.02 of the lowest item by store on every data set. They also show why one window is not enough: on M5 the official test month was the only one of four in which LightGBM was the most accurate over all levels (0.547 against 0.638 for the world model), so a comparison on that month alone would have ranked it first.

10 What was tried and dropped

Table 19 lists what this revision tried and did not keep; Table 31 in Supplement C lists what the first version tried, with its failures. Supplement B.4 scores every change to a model setting with saved runs against the configuration it changed (Table 26), every change to the loss (Table 4), and lists every other result in the record (Table 27). Some changes that were kept won half their comparisons or fewer: the 8-week item level on Dominick's was worse than the 26-week level in 15 of 28 categories and was kept because it removed the failures that decided the mean, and the normalised roll-forward was worse in three of six Dominick's categories at one seed and was kept for the steadiness it gave with averaging.

The forecast after a promotion. The persistence test showed the world model carrying a promotion forward as further lift where shoppers stock up. Studies C and E tried to give the dip a route. A stock term, one number per item that builds with sales above the item's level, decays by a learnt share and lowers later sales by a learnt amount that cannot be negative, made the world model worse overall in three of six categories on the development windows. On the lift model it passed the development rule only on the mean of the runs: the repeat on the second runtime alone was worse overall than the world model in four categories and would have failed. On the rolling windows it did not fix the after-promotion forecast (soft drinks 1.27 with it, 1.28 for the plain lift model, 1.19 for the world model), it made the snack-cracker total worse than the plain lift model's (1.510 against 1.026), and its total error was higher than LightGBM's in all

Table 19: What the revision tried and did not keep. Each was written down with its pass rule before it ran. dev.: the development windows of the six Dominick's categories. After promotion: forecast over actual in the two periods after a promotion.

Change or model	Where tested	Result
Stock term on the world model (study C)	Dominick's dev., seed 1	Overall worse than the world model in three of six categories
Plan of the two periods before, on the lift model (study C)	Dominick's dev., seed 1	Overall worse than the world model in three of six
Stock term on the lift model (study C)	Dominick's dev., then rolling	Passed on development on the mean of the runs only; the second-runtime repeat alone failed (after promotion 0.97 soft drinks, 0.93 snack crackers); on the rolling windows 1.27 and 1.16, against 1.19 and 1.17 for the world model
Loss weight $1 + w$, $w = 2$, after a promotion (study E)	Dominick's dev., seed 1	Error after promotion lower in all six (soft drinks 0.174 to 0.150); overall worse in four (snack crackers 0.749 to 0.857)
Loss weight $1 + w$, $w = 5$ (study E)	Dominick's dev., seed 1	Overall worse in four; snack crackers after promotion 1.11
Fitted ensemble weights (study A)	All nine, rolling	Better than equal weights in five of 54 comparisons, worse in 13
Timestral in place of TimesFM (study F)	All four sources, rolling	Worse on Dominick's and Frat; better at the M5 total only
Timestral-209M (study F)	Frat, rolling	Less accurate than 28M; too slow for the other data sets
MinT, shrunk covariance, as first run	Frat, rolling	Total WRMSSE 247.170 from a singular covariance; after the fix 0.610

six categories. A loss weight on the periods after a promotion lowered the error there in all six categories and made the overall score worse in four. None became a default. The reason given for study C also needed a correction: the 1.32 and 1.33 after-promotion ratios that prompted it came from an earlier world-model version; the final world model was at 1.00 and 1.04 on those windows.

Seeds and run-to-run noise. Stage 2 of study C repeated each configuration with seeds 2 and 3 and got identical scores in all twelve runs. Single-fit configurations whose settings named no seed trained with seed 0 for every run seed; the seed now reaches the model, and the results already recorded are those of seed 0 (each manifest records its commit). The repeat also showed that two runs of the same configuration on different Colab machines differ by as much as the variants being tested: the world model's overall score in snack crackers was 0.911 on one machine and 0.749 on another. Study E therefore reran the world model in the same job as the variants it was compared with. The rolling windows, being one fit per model, carry this noise; the paired tests over windows absorb part of it, not all.

11 Discussion

Table 20 maps the rule of Section 8.5 to the decisions a retailer makes from the forecast. Each row is what the tuned models did on the nine data sets studied, within the scope of Section 12.

Why the average works. The two tuned models are close in accuracy and wrong in different places. Across the nine data sets, each is clearly more accurate than the other in a similar number of paired comparisons (64 for the world model, 75 for LightGBM), and the data sets on which each leads differ. An equal-weight average of two forecasts whose errors differ is more accurate than either on most series, and no weights have to be estimated, which the forecast-combination literature finds is hard to beat [68, 69].

What the world model brings. The world model's contribution is clearest where the forecast has least to go on. On items it was not trained on it lost at most 0.040 in scaled error, against up to 0.311 for LightGBM, and forecast within 11% of actual sales; LightGBM forecast as little as two thirds. It reads an item's recent sales into a state and carries that state forward under the plan, so it does not need the item in its training data, whereas LightGBM takes the item as one of its labels and has learned nothing for an item it was not trained on. The same carried state keeps its totals over the promotion block closer to actual sales than LightGBM's on M5, Favorita and soft drinks.

What the trees bring. LightGBM is the more accurate of the two on frozen entrees and snacks at every grain, and its block totals are closer on analgesics, bottled juice and snacks. It needs no pretraining and trains in minutes, and on lines with history it is as accurate as the world model on most data sets. On snacks, where the world model alone is weakest (1.009 overall against 0.865 for LightGBM), the average stays close to LightGBM at 0.884.

Table 20: The decisions a retailer makes from one item-by-store forecast, and what the tuned models did at each on the nine data sets studied.

Decision	Grain	Forecast to use	What the tuned models did
Store orders and inventory	Item by store, by day or week	The average	Lowest error at the item levels on all nine data sets; clearly less accurate than either model in at most 8 of 100 paired comparisons
Promotion and supplier funding	Item across stores, by week and over the promotion block	The average	Lowest at the item levels on all nine; over the whole block lowest on five and second on the other four; LightGBM alone drifts in total on M5, Favorita and soft drinks
Category performance	Category by store and category, by week	The average	Lowest at store grain and above on eight of nine; LightGBM lowest on snacks
Financial plan	Chain total	The average	As for category performance; the total is the sum of the same item-by-store forecast, so it agrees with every plan below it
New items and range changes	Item by store, items not yet trained on	The world model alone	Clearly more accurate than the average on all five data sets tested; LightGBM forecast 66 to 93% of actual sales on these items
New stores	Item by store, stores not yet trained on	The average	The average and the world model each clearly ahead on one data set and level on three; LightGBM clearly behind the world model on four of five
Choosing for another retailer	Any	Rerun the comparison	The order of models is a finding on these data; every comparison, including those that lost here, should be rerun on the retailer's own data

12 Limitations and threats to validity

Scope: two of the four Ps, on public data. The paper addresses price and promotion. It forecasts items newly added to a range but does not decide product (which items to range) or place (space, distribution), and it does not solve the four Ps together; a unified forecast across them does not exist in this work, and one could change which model works best at each grain. The data are four public data sets with their own plan inputs and constraints. A single retailer's data, with its own levers, stock and space records, could give a different ordering of models, and the grain comparison should be repeated on it before a choice is made.

Adapting the findings. The rule of Section 8.5 and the ordering of models behind it hold on the nine data sets studied. They are not a ranking of methods for retail in general. A retailer or researcher adopting the framework should rerun every comparison on its own data, including the models and combinations that lost here, since data with other levers, lines and lengths of history can change the order. The code and settings for every comparison are released for that purpose.

Forecast horizon. Every forecast in the study runs 8 weeks ahead on the weekly data and 28 days ahead on the daily data. Whether the models' accuracy, and the order between them, holds over the 26- and 52-week horizons of range, space and supplier reviews was not tested: a longer forecast may degrade, and may degrade differently for each model. Testing it needs a retailer's own plans and range decisions to score against and is left for separate work.

New items. The kept-out items of the transfer test were not trained on, but each model read their recent sales before forecasting them. An item with no sales at all, on its first day in range, was not tested; the new-item part of the rule covers items the models have not been trained on, not items with no history. That part of the rule was drawn from the results, after a wider rule written down in advance held only in part (Section 8.5).

Windows and seeds. Trained models are one fit each in the rolling windows, and runs of one configuration on different machines differed by as much as some differences tested (§10). Four to six windows per data set is a thin basis for a bootstrap over windows and for a model confidence set: a bootstrap over six windows can draw few distinct samples, its intervals are approximate, and the set has little power, so a model kept in the set has not been shown to be as good as the best. Adjacent windows share training history and season, so they carry less information than their number suggests. The item bootstrap within a window treats the items as the sample and does not cover variation between windows. Single-fit runs before 16 October trained with seed 0 whatever seed was named.

Studies added after the main results. Studies A to G were added after the rolling results were read. Each was

written down before it ran, but which studies to run was chosen knowing that the ensemble came first. Timestral was run at 28M parameters, not 209M, because the larger model was too slow, a change made before any Timestral result.

Comparison models. TFT was not run on M5 or Favorita, and Chronos-Bolt and TimesFM from history alone not on Favorita. TimesFM with the plan fits its linear plan model within groups of 2,048 series. Pretrained models are used as published with their point forecast fixed in advance (the mean), which for Timestral let a few extreme upper quantiles dominate on weekly data. The Timestral tail check and the cheese window were looked at after the results were read.

What lift means. Merchants promote in weeks they expect to sell well, so part of a measured lift is good timing. The lift is what the forecast attributes to the plan, not what the plan caused. Our companion audit [67] measures the gap in generated store data where the true lifts are known: the world model's and LightGBM's lifts were a third too large while every check against the model passed. Plan responses in this paper, including the perturbation tests, are forecasts, not causal estimates.

What the data do not record. None of the data sets records stock, so a sell-out looks like low demand; or shelf space. Favorita records no prices. The data were collected between 1989 and 2017 and their plan inputs differ, so the results compare models within each data set. The decision costs are simulations under stated costs.

13 Conclusion

A retailer makes decisions from the same demand at different grains, and on these data no single model was the most accurate at every grain. One forecast made item by store and summed upward, as the equal-weight average of a tuned world model and tuned gradient-boosted trees, was the most accurate choice at store grain and above on eight of nine data sets and at the item levels on all nine, on the fast lines and on the slow tail. For items the models had not been trained on, the world model alone was more accurate than the average on every data set tested, because LightGBM under-forecast them. Settings carried from one retailer's data to another's, and training on other retailers' data did not help. The rule that follows, the average everywhere and the world model alone for new items, is a finding on these data for forecasts of 8 weeks and 28 days under price and promotion plans. The comparison behind it, run over several windows with paired tests and every losing model reported, is what we propose a retailer reruns on its own data, and over its own planning horizons, before adopting it.

14 Reproducibility and compute

Records. All code, configuration files and tests are in the repository. A run refuses to start on uncommitted code and writes a manifest holding the code commit, the full configuration and its checksum, the seed, checksums of every data file read, package versions, the machine and run times. The manifest and scores of the 110 final-window runs are committed under `results/promoted/`. The revision's analyses of saved forecasts are committed as tables under `results/paper1_revision/tables/`, with the per-window scores of study F under `results/paper1_revision/scores/`; the plan and every amendment are in `docs/PAPER1_REVISION_PLAN.md`. `make` in `paper/` rebuilds every figure, table and quoted number from these files (26 seconds from a clean copy of the repository on two CPU cores, 17 October 2026), and a quoted revision number that is missing from them stops the build. One table was transcribed from a console run rather than written by a script: Timestral's quantile sums (`studyF_tails.csv`), from `scripts/timestral_tails.py`. Forecast files and model checkpoints are kept outside the repository with their checksums recorded.

Compute. Every run used one Google Colab machine: an NVIDIA Tesla T4 with 14.6 GB of memory, a 2-core Intel Xeon CPU @ 2.00GHz and 12.7 GB of memory, PyTorch 2.11.0+cu130 with CUDA 13.0. Table 21 gives the run time of every timed final and comparison run of the first version: 94 runs and 44.7 machine-hours. The revision's rolling windows, studies and analyses were not timed as a whole; from the working sessions we estimate about 60 further machine-hours on T4 and L4 GPUs, of which study F took about 12. This is an estimate, not a record.

15 Data, code and licence

Data. M5 is available from the competition's Kaggle page [1]; Dominick's from the Kilts Center for Marketing, University of Chicago Booth School of Business [19]; Breakfast at the Frat from dunnhumby [20]; Corporación Favorita from its Kaggle competition [71]. Each is obtained from its provider under the provider's own terms. No data is redistributed with the code.

Table 21: Run times of the final and comparison runs, from their manifests. Wall-clock minutes per run (data loading, training, forecasting and scoring) and hours in total, on one Google Colab machine with an NVIDIA Tesla T4 (LightGBM uses its CPU). Dominick's: six categories together; a Dominick's world-model run averages three fits. Two M5 runs resumed after the Colab runtime dropped record only the resumed part (the shortest values in their rows). Not counted: runs saved before run times were recorded, the development study, and TimesFM on Dominick's and Frat.

Data	Model	Runs	Median	Range	Hours
M5	World model	9	101.5	37.3 to 103.2	14.0
M5	Lift model	9	88.7	10.3 to 94.6	12.2
M5	LightGBM	9	78.1	76.5 to 79.9	11.7
M5	Graph model	12	7.2	7.2 to 7.5	1.5
M5	TimesFM	8	3.9	3.8 to 4.1	0.5
M5	Seasonal naive	2	1.3	1.2 to 1.3	0.0
Dominick's	World model	18	9.2	9.0 to 10.6	2.8
Dominick's	LightGBM	18	4.6	3.1 to 11.5	1.7
Frat	World model	3	2.2	2.2 to 2.2	0.1
Frat	Lift model	3	2.0	1.9 to 2.0	0.1
Frat	LightGBM	3	1.5	1.5 to 1.6	0.1
All 94 timed runs					44.7

Code. The code is at `https://github.com/karthik-iyer-ad/retail-world-models`; the version used for this paper is archived as release v1.0.0 on Zenodo, doi:10.5281/zenodo.23448056.

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16 Author contributions

Karthik Iyer conceived and scoped the work, designed the study, wrote the code, ran the experiments and analyses, and wrote the paper, with AI assistance as declared in §18. Madan Dabbeeru widened the range of methods considered for each research question, and reviewed the work and the manuscript.

17 Practice background and independence

The first author has worked in retail for fifteen years, across continents, in grocery, department stores, fashion, electronics and online marketplaces. No data, method, code or other intellectual property from those roles or engagements is used in this work, and no employer or client data are used: the study is built from the ground up on public data sets. What the authors carry from practice is the set of questions retailers ask of a forecast, which shaped the research questions in Section 2. The first author's research focus is world models; the second author widened the scientific options considered for answering each question.

18 Declaration of generative AI and AI-assisted technologies

The research questions, scope, study design, choice of data and methods, and interpretation of results are the authors' own. Claude (Anthropic) was used as an assistant to write and test code, run analyses, produce figures and tables, and draft and edit text. The authors directed this work throughout, reviewed every result, and corrected and revised the code, analyses and text over many iterations, and take full responsibility for the content of the publication.

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A Constraints the models work within

Table 22 lists the constraints. The world model's settings are in Table 3.

B Supplement: the development study

This section reports what was tried before the final tests, on the development windows only. It shows how the final models were chosen and which ideas from the literature did not help on these data. Every row is a section of the results record, with its run folders.

B.1 Breakfast at the Frat

Benchmarks. On the three windows before the data set's end, seasonal naive scores 0.832, the average of the last 8 weeks 0.756, LightGBM with price only 0.554 and LightGBM with every recorded lever 0.447. The levers are worth a fifth of the error: display, feature and price reduction carry information price does not.

What helped and what did not. Figure 16 shows the configurations of the main comparisons on Frat; Table 27 lists the eleven other Frat configurations screened, including the worst (0.458) and the best (0.400). Our item encoder with a direct readout scored 0.424, level with LightGBM (0.421). Building the forecast as a baseline under the nothing-planned plan times a lift (the lift model) brought it to 0.406, better than LightGBM in all five runs and with half the spread across seeds. Letting the 55 items in a store affect each other, by conditional similarity, attention or both, made the forecast worse in every version (0.443 to 0.476) and less steady; limiting it to items in the same category narrowed the gap without closing it. Extra inputs did not help: the plan of the four weeks before (0.411), the same week last year (0.408), the US seasonal calendar (0.425), the count of items on promotion in the store (0.406) and the recorded base price used as the regular price (0.434). The recorded base price changes about twelve times a year per item and equals the shelf price in every week without a promotion, so it is not a slow-moving regular price.

Latent (JEPA-style) losses. Following the JEPA literature [49, 50] and M-JEPA [61], we added a loss that predicts the encoder's own state for the forecast weeks from the state now and the plan, with a slowly updated copy of the encoder as the target [52]. No recipe beat the lift model: alongside the forecast loss 0.410 to 0.411, pretrained on the latent loss and then fine-tuned 0.407 to 0.414. With the encoder frozen after latent-only pretraining the model still reached 0.428, so a state learned without any sales target carries most of the forecast signal. Long pretraining followed by a slowed encoder gave the best display-week error of any configuration (0.813 for the world model, against 0.817 for LightGBM).

The world model. Rolling the state forward week by week under the plan scored 0.408, level with the lift model on the overall score and on every lever. Three ideas from the world-model literature, each at one untuned setting, did not help: keeping the plan's effect apart from the no-plan state as in DWM [57] (0.417), a latent check on each rolled step with a VICReg penalty [53] and near weeks weighted more as in TD-MPC2 [56] (0.427), and all three together (0.424). Training changes did not help either: stopping at the best point on the last 8 training weeks (0.425) took 8 of only 90 training weeks out of the targets, and a term on summed totals in the loss (0.414) had no bias to fix, since total forecasts were within 3% of sales in every run. Readout dropout with weight decay (0.409) was level and steadier, and was kept for the lift model.

Beyond accuracy. Table 23 reports four further tests. *New items*: with one product in ten held out of training, the lift model borrowing the state and level of the ten known items in the same store with the most labels in common scored 0.890 on the new items, against 1.310 for LightGBM fitted on labels, price and plan, and 1.510 for a forecast of zero; hiding the item label in a tenth of training examples changed this little (0.879). On Dominick's analgesics, where items carry no category or maker labels, borrowing failed (2.999 against 0.923): the method needs item attributes. *Combination*: weighting the lift model and LightGBM on a validation window gave the best Frat result of the study (0.388), better than either part on display and circular weeks, because the two make different errors in promoted weeks; on analgesics the combination was level. Combinations were not carried to the final tests, which compare single models. *Plan A against plan B*: for pairs of weeks of the same item and store with different plans, every model picked the week that sold more about 78% of the time; LightGBM was marginally ahead. *Unusual plans*: the test as first defined, a lever mix under 2% of all training rows, found no such week in the windows, so it was redefined after that run as a mix the item itself had run in under 2% of its weeks. On those, no model degraded much and LightGBM stayed ahead. The world model's ability to roll a plan forward did not show as an advantage on these tests.

B.2 Dominick's

Analgesics, the category used to set the models. On analgesics, seasonal naive scored 1.274, the average of

Table 22: Every constraint the trained models and the ensemble work within, with its value and where it lives in the code.

Constraint	Value	Where
Log forecast bound (world and lift model)	$[-10, 10]$ around the item's level	`src/rwm/model/state_model.py` readout
Lift when nothing is planned	Exactly zero	Readout; tested in `tests/`
Baseline under a changed plan	Unchanged	Readout; tested in `tests/`
Regular price	Highest price in the last 12 periods	Plan features
Brand-switching share α	One per run, fitted on the last 8 training weeks	Dominick's world model
Forecasts	Not below zero	Every model
Ensemble weights	One third each, fixed 13 October 2026	`scripts/revision_analysis.py`
Interval pooling	Horizons pooled below 50 earlier errors per horizon	`src/rwm/evaluation/compare.py`
Interval ends	Rounded outwards to whole units (not Favorita)	`src/rwm/evaluation/compare.py`
TimesFM covariate fit	Groups of 2,048 series	`src/rwm/prior_work/`
Timestral context and batch	512 periods; batch 256, halved on running out of GPU memory	`src/rwm/prior_work/timestral_zero_shot.py`

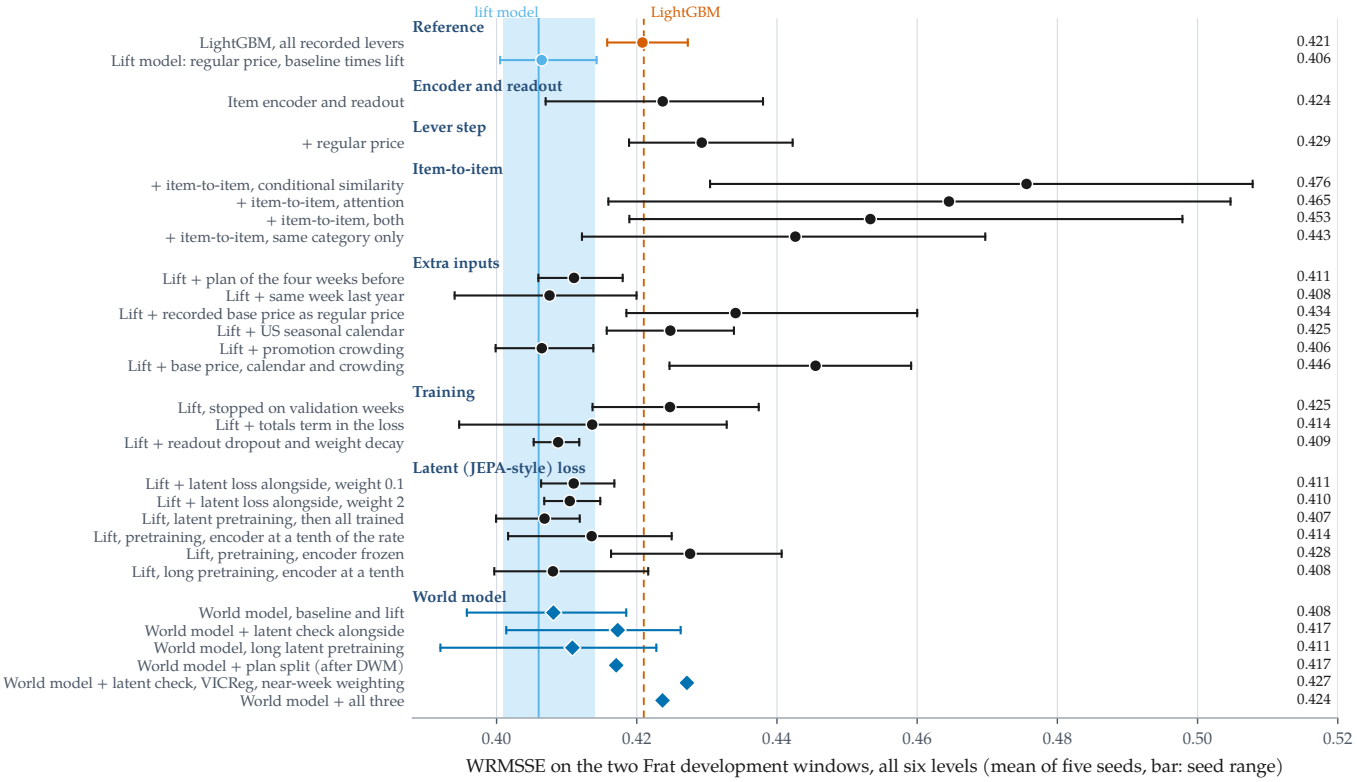


Figure 16: On Frat the gain came from building the forecast as a baseline times a plan lift; item-to-item links, extra inputs, latent losses and the roll-forward did not improve on it. Each row is one configuration on the two development windows, five seeds: mean (marker) and range (bar). Blue band: the lift model's seed range; vertical lines: the lift model (0.406) and LightGBM (0.421). Values below the lift model are in bold. Rows without a bar were recorded as a mean only.

the last 8 weeks 0.723 and LightGBM 0.715. The lift model scored 0.676 on three seeds, and the world model 0.667 once a totals term was removed from its loss: that term made the total worse, pulling forecasts 8% low (0.819 for the lift model). Training four times longer without a stop broke the summed levels (1.473) while item-by-store error barely moved, so a small bias in the same direction on every item added up at the category total. The seasonal calendar with promotion crowding made LightGBM worse (0.715 to 0.734) and, with a held price as the regular price, took the lift model to 0.919 by the same route; the held price alone was level (0.674). Item-to-item links did not help (0.697). Of the world-model ideas that failed on Frat, the plan split improved item-by-store error (0.707 to 0.693) and broke the total (0.687 to 0.793), and all three together scored 0.730

Table 23: What the plan-aware models were also tested on during development: new items, combination with LightGBM, choosing between two plans, and plans an item had rarely run. Development windows, three seeds (Frat combination: five). WRMSSE, lower is better, unless stated. Plan A against plan B: share of week pairs where the forecast picks the week that sold more, and the average miss on the size of the change (log scale).

Data	Model	Result
New items: one product in ten never seen in training (item by store)		
Frat	LightGBM, new item from labels, price and plan	1.310
Frat	Lift model, borrowing from similar items	**0.890**
Frat	Forecast of zero	1.510
Analgesics	LightGBM, from labels, price and plan	**0.923**
Analgesics	Lift model, borrowing (no attribute labels)	2.999
Combination with LightGBM, weights set on a validation window (all levels)		
Frat	Lift model + LightGBM	**0.388**
Frat	World model + LightGBM	0.390
Frat	LightGBM; lift model; world model	0.421; 0.406; 0.408
Analgesics	Lift + LightGBM; world + LightGBM	0.670; 0.668
Plan A against plan B: same item and store, two weeks with different plans (Frat)		
Frat	LightGBM	78.1% right; miss 0.448
Frat	Lift model	77.4% right; miss 0.460
Frat	World model	77.5% right; miss 0.459
Frat	World model + plan split	77.5% right; miss 0.458
Unusual plans: a lever mix the item had in under 2% of its weeks (item by store)		
Frat	LightGBM	0.696 unusual; 0.697 usual
Frat	Lift model	0.742 unusual; 0.725 usual
Frat	World model	0.730 unusual; 0.730 usual

against 0.667. Five seeds changed the picture: the world model held (0.666) and LightGBM scored 0.702, but the lift model fell to 0.718, so a three-seed figure had flattered it.

All 28 categories. Run on all 28 categories with settings from analgesics, LightGBM was the most accurate of the three in 14, the lift model in 5 and the world model in 9, with mean WRMSSE 0.688, 0.784 and 0.747; on promoted weeks LightGBM was most accurate in 24 of 27 (Figure 17, Table 24). Analgesics did not generalise, which is why the paper reports every category. The failures (snack crackers 1.463, bath soap 1.465, the lift model on soft drinks 1.406) sat at the category total: item-by-store error was close to LightGBM's. On a six-store sample of snack crackers (one seed, 800 training steps, recorded in `docs/PAPER_NOTES.md`) the cause was in-and-out and seasonal lines: items recorded in 8 to 20 of the last 26 weeks were 8% of sales and the lift model forecast them at 9 times their sales, because the model set each item's level from its 26-week average after the item had fallen away. Capping extreme weeks did little; taking the level from the last 8 weeks removed most of it. On all 28 categories the 8-week level brought the world model's mean from 0.747 to 0.717 by removing the large failures (snack crackers 1.463 to 0.874, bath soap 1.465 to 1.195), but it was worse than the 26-week level in 15 of the 28 categories (cheese 0.996 to 1.116, cereals 0.673 to 0.805), the median rose from 0.689 to 0.736, and it beat LightGBM in 10 categories against 14 before. It became the Dominick's default because it removed the failures that decided the mean, not because it was better in most categories. On eight categories the 8-week level beat a 4-week level in five for the world model, and the 4-week level was better in five for the lift model, which shares the setting; the lift model with an 8-week level broke bath soap (1.568 to 2.203), and with a 4-week level cigarettes (0.918 to 1.245). LightGBM, which sees short recent averages directly, does not have the problem.

Peaks and dips. In cheese, both our model and LightGBM under-forecast promoted weeks by about a third in the holiday window, and cheese peaks do not sit on a fixed week relative to the holidays: they follow the chain's promotion timing, which event flags cannot place. Figure 18 splits forecast against sales into promoted weeks, the weeks just after a promotion, and ordinary weeks, for six categories. The world model did not see the dip after a promotion where the dip is large: it forecast 1.33 times sales after a promotion in snack crackers and 1.32 in soft drinks, against 0.92 and 0.90 for LightGBM. Feeding the model's own sales forecast

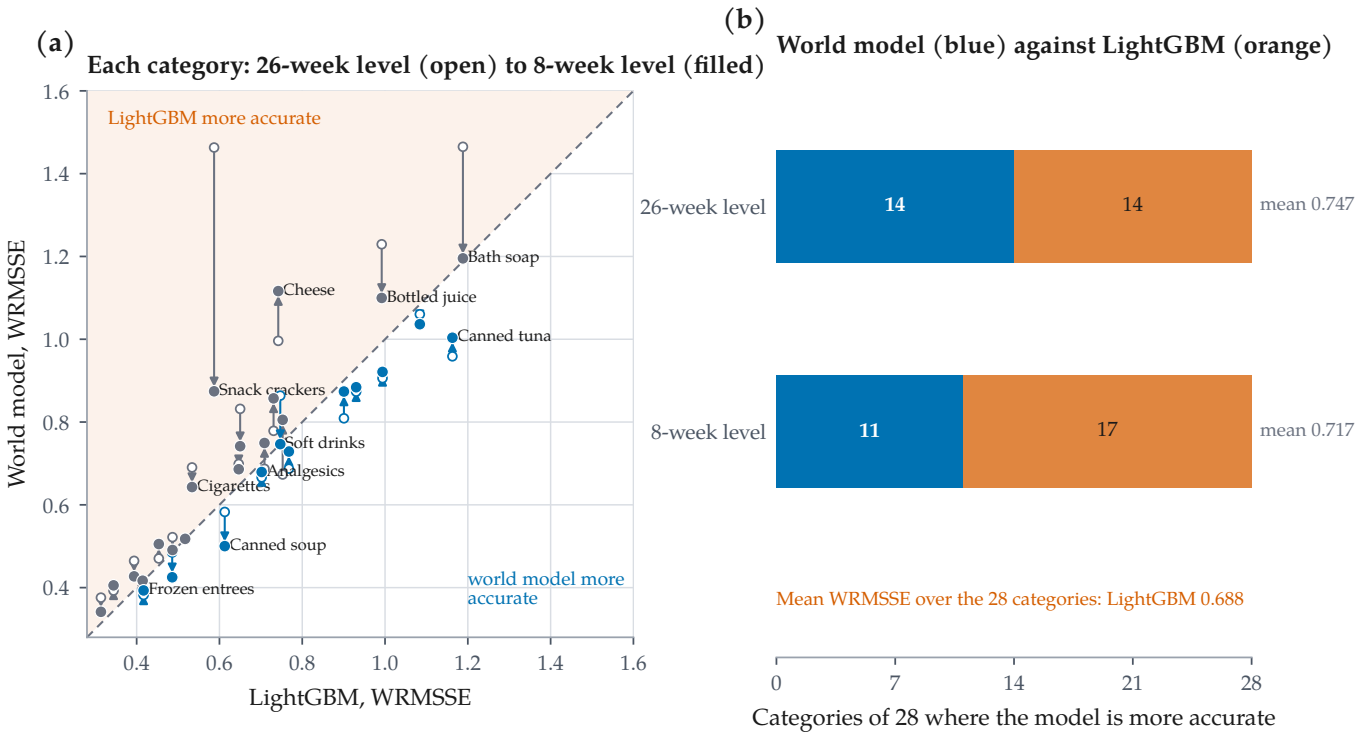


Figure 17: Run unchanged on all 28 Dominick's categories, the world model set on analgesics failed badly in a few; taking each item's level from its last 8 weeks removed the worst failures but did not make it better than LightGBM on average. Development windows, one seed. **(a)** Each category's WRMSSE for LightGBM (horizontal) and the world model (vertical), with the item level from the last 26 weeks (open) and the last 8 weeks (filled); blue: the world model is more accurate with the 8-week level. **(b)** Categories where each model is more accurate, and the mean over the 28.

into the roll-forward, so that a promotion's sales could shape the weeks after it, fixed the soft-drink dip (1.32 to 1.00) and made the overall score worse in four of six categories. Weighting the loss by dollar sales and by how far a week sits from the item's usual level fixed cheese promoted weeks (1.21 to 1.00) and was worse overall in three categories. Both together were worse overall than the world model in four categories (snack crackers 0.874 to 1.294, bottled juice 1.100 to 1.222, analgesics 0.679 to 0.901) and the best of all models only in frozen entrees (0.377). Adding the normalised roll-forward to the sales feedback put the snack-cracker forecast after a promotion right (0.97 of sales) and broke its category total (1.60; overall 1.268). In frozen entrees every model, LightGBM included, forecast about 0.7 of sales after a promotion: there the week after is stronger than any model expected.

From the first world model to the one tested. Figure 19 follows the world model on the six categories through the changes kept, three seeds each. Across seeds the first version was unsteady: its range within a category was 0.04 to 0.38, against under 0.03 for LightGBM, and on the mean LightGBM was ahead in five of six. Brand switching fixed what it was built for: in snack crackers and soft drinks, where promoted weeks broke the total, it was better on every seed (0.813 to 0.752, 0.902 to 0.789) and item-by-store error improved too. Elsewhere it did not help: it was better on no seed in cheese and frozen entrees and on one of three in bottled juice and analgesics, and the mean was worse in cheese (1.088 to 1.111) and frozen entrees (0.406 to 0.416). It was kept for all six. The fitted share of the summed lift that the category total keeps (Figure 19g) is 0.00 to 0.66 in snack crackers, against about 0.9 in bottled juice, analgesics and frozen entrees: in snack crackers the summed item lifts overstated the category total, and the fit took most of it back. Where it comes out at 1.0 the step changes nothing, a built-in check. The direction agrees with van Heerde, Gupta and Wittink's household estimates [15], but the share is fitted to make totals right; it is not an estimate of switching. Normalising the rolled state, tried first on one seed, was better in two categories, worse in three (analgesics 0.679 to 0.790, soft drinks 0.747 to 0.882) and level in one, and the record ruled it out as a general fix. With switching and three seeds it narrowed the seed range in five of six categories, but the mean was worse than switching alone in snack crackers (0.752 to 0.830) and soft drinks (0.789 to 0.827). Added to switching with three fits averaged (seed 1), it was better in four categories and worse in cheese (0.921 to 0.983) and snack crackers (0.722 to 1.806); three fits averaged without it

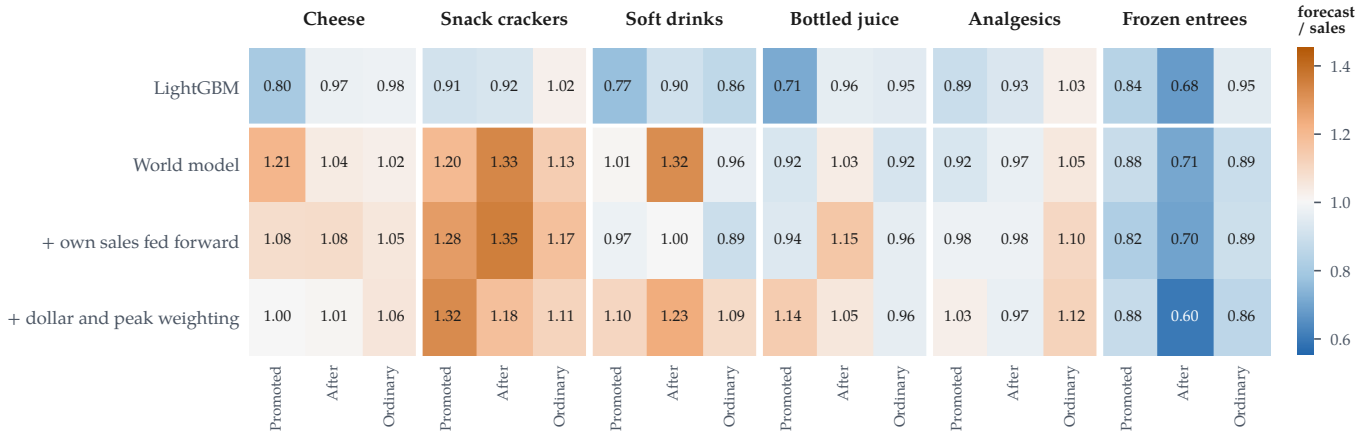


Figure 18: The world model over-forecast the weeks after a promotion where shoppers had stocked up; the two fixes each repaired one category and hurt others. Forecast against sales (1.0 is right; orange over, blue under) in promoted weeks, the weeks just after a promotion and ordinary weeks, Dominick's development windows, one seed. Rows: LightGBM, the world model with the 8-week level, with its own sales forecast fed forward, and with the loss weighted by dollar sales and by peaks.

had broken bottled juice (1.170). It was kept because the combination was the steadiest version. Averaging three fits per run with normalisation gave the steadiest version (range 0.05 or less in five categories) and the candidate for the final test: better than LightGBM on the three-seed mean in analgesics (0.621 against 0.701), frozen entrees (0.391 against 0.419) and soft drinks (0.743 against 0.759). One snack-cracker run of the three failed (1.806): its second window's category total broke with promoted weeks forecast at 2.6 times sales while item-by-store error stayed normal, so the summed promotion lift can still run away on one fit in three. Twice the training steps, screened after the final tests were fixed, was better in two categories, worse in two, level in one and mixed in one, and was not kept.

Other changes. Showing the model both its 26-week and its recent levels as inputs, instead of setting its scale by the recent level, was the best world-model version in four of ten categories but did not fix snack crackers (1.515) or soft drinks (1.069): the model did not learn to switch to the recent level when the long level still set its scale. Item role labels (new line, in-and-out, volume driver, long tail, high-low, promotion-responsive, core), assigned from each item's own sales before the windows, moved LightGBM by at most 0.013, since it reads the same information from its history, and moved the world model in both directions; letting the role pick the level broke snack crackers (2.422). We did not call the volume drivers key value items, since a key-value-item list rests on shopper price perception, which these data do not record.

B.3 M5

On the M5 practice window (Table 25), the item encoder with a direct readout scored 0.699 over five seeds against 0.715 for LightGBM, and the lift and world models were level with LightGBM over three seeds (0.713 and 0.707). Normalising the rolled state was the one change that separated the world model: 0.656, with its worst seed (0.675) better than the best seed without it (0.679), all of the gain at the totals and the seed range cut from 0.071 to 0.029. On Dominick's 8 weekly steps it was worse in three of six categories at one seed (§B.2), which fits the reading that it stops the state drifting over a long roll and has little to stop over a short one. Averaging three fits did not help on M5 (0.665), where single fits were already steady.

After the official test month, an error analysis of the world model found its 2.6% excess in Foods (+3.9%), in one store (CA_3, +7.5%), and on Sundays and Mondays (+5.1%, +6.4%), not growing with the horizon. Grouped by sales in the 28 days before the test, the excess came from items that had been selling fast (+5.4% and +6.1% on items selling 3 to 10 and over 10 a day). Four changes were screened on the practice window under rules written before screening: weekday as seven inputs, event kinds, twice the training steps, and the item's level against its past-year level. None beat the original (Table 25), so there was no second version. On the practice window every version under-forecast by 4% to 6%, while on the test month the world model over-forecast by 2.6%: the sign of the level error changed between adjacent months, so a fix tuned on one would not carry to the next. That finding is what led to the confirmation months of §9.10.

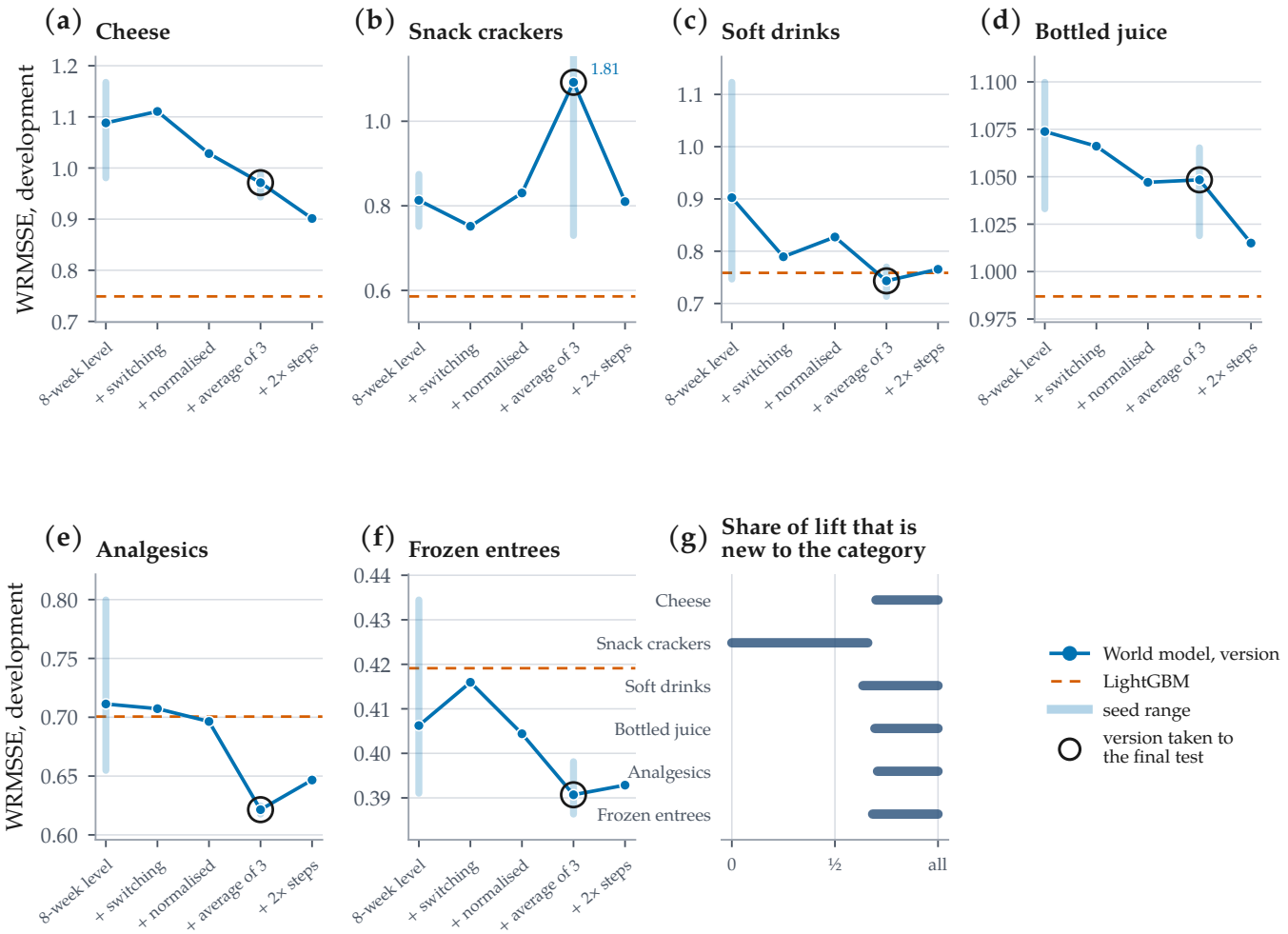


Figure 19: Brand switching, a normalised roll-forward and averaging three fits took the world model past LightGBM in three of six categories on the development windows, and the switching step learned how much of a promotion's lift is new to the category. **(a–f)** Development WRMSSE of each version, three-seed means; shaded bars: seed range of the first version and of the one taken to the final test (circled); dashed: LightGBM. **(g)** The fitted share of the summed promotion lift that the category total keeps, the split that makes category totals fit best (not an estimate of switching), range over runs.

B.4 Every change and run in the record

Every run is recorded in `results/README.md`, with its configuration and run folders. Table 26 scores every change to a model setting that has saved runs against the configuration it changed: 23 changes over 157 comparisons. Several changes that were kept won about half their comparisons or fewer; the text above gives the reason each was kept. Table 27 lists the results in the record not shown in the figures and tables above, and the configurations that were written and never run on full data.

C Supplement: the final windows of the first version

The first version of this paper scored each data set once, on its final window, with three seeds per trained model. These results come before the rolling windows of the main text and are kept here for the record.

The first version scored each data set once, on its final window, and its findings stand: on M5 the most accurate model depended on the level and the month (Table 28), and on Dominick's and Frat on the category (Table 29). On M5, TimesFM given only history was the most accurate single model item by store in all four months (0.885 on the test month, against 0.897 for the world model and 0.905 for LightGBM), while the plan-aware models were far more accurate at the totals. The official test month was the only one of four in which LightGBM was the most accurate over all levels (0.547 against 0.638), so a comparison on that month alone would have ranked it first. Run on the final windows (Table 30), the ensemble is lowest or within 0.02 of the lowest item by store on every data set, which agrees with the rolling windows. One final run did not finish as first started: the frozen-entrees world model, seed 3, stopped when the

Table 24: Across all 28 Dominick's categories LightGBM is more accurate on average; the world model wins in 14 with its first level and 10 with the 8-week level. Development windows, one seed (analgesics: five), settings set on analgesics and not tuned per category. World model with the item's level from its last 26 weeks or last 8 weeks. Lowest in each row in bold.

Category	LightGBM	Lift	World, 26 wk	World, 8 wk
Analgesics	0.702	0.718	**0.666**	0.679
Bath soap	**1.188**	1.568	1.465	1.195
Beer	**0.454**	0.460	0.470	0.505
Bottled juice	**0.992**	1.247	1.229	1.100
Canned soup	0.613	0.532	0.583	**0.500**
Canned tuna	1.163	1.030	**0.959**	1.004
Cereals	0.753	0.795	**0.673**	0.805
Cheese	**0.742**	0.900	0.996	1.116
Cigarettes	**0.534**	0.918	0.690	0.643
Cookies	**0.487**	0.503	0.522	0.491
Crackers	0.709	0.967	**0.687**	0.749
Dish detergent	**0.344**	0.436	0.393	0.405
Fabric softener	0.414	**0.358**	0.397	0.417
Front-end candies	**0.650**	0.748	0.832	0.742
Frozen dinners	0.930	0.905	**0.873**	0.884
Frozen entrees	0.417	**0.374**	0.382	0.393
Frozen juices	**0.314**	0.360	0.376	0.342
Grooming products	0.901	1.208	**0.809**	0.874
Laundry detergent	0.768	0.735	**0.687**	0.729
Oatmeal	1.084	**1.026**	1.060	1.036
Paper towels	0.486	0.445	0.485	**0.425**
Shampoos	0.994	0.938	**0.906**	0.921
Snack crackers	**0.587**	0.959	1.463	0.874
Soaps	0.517	0.521	**0.516**	0.518
Soft drinks	0.747	1.406	0.864	**0.747**
Toilet tissue	**0.394**	0.413	0.465	0.427
Toothbrushes	**0.647**	0.671	0.699	0.686
Toothpaste	**0.731**	0.802	0.779	0.857
Mean	0.688	0.784	0.747	0.717

runtime dropped, could not be resumed because of a fault in restoring its random state (since fixed), and was run again from the start with the same settings (0.476 overall).

How much the plan matters. On M5, where prices rarely change, giving TimesFM the plan changed its item-by-store error by at most 0.002 in any month. On Dominick's, where promoted weeks carry 18% to 61% of a category's units, TimesFM without the plan forecast 29% to 63% of promoted-week sales and with it 60% to 113% (Figure 21a). The trained plan-aware models also fell short in promoted weeks: the world model forecast 48% to 88% of promoted-week sales and LightGBM 50% to 84%.

First-time price cuts. In the April M5 window, HOUSEHOLD_1_146 had sold almost nothing for a year at 5.98, then sold 120 units in four stores over a window that held a one-week cut to 4.50 (Figure 21b). Models working from history forecast almost nothing: LightGBM 3.0, 3.0, 3.2 units across seeds, TimesFM 7.8. The world model, which reads the cut through its plan, forecast 121, 600, 243; the graph model 59, 45, 55; the lift model 79, 206,886, 316. How large the response should be is not fixed by the data: the world-model seeds differ by a factor of five, and one lift-model seed forecast over 200,000 units.

Table 25: On the M5 practice window, only the normalised roll-forward separated the world model from LightGBM; no later change beat it. WRMSSE over all 12 levels, 25 April to 22 May 2016. A screened change was kept only if it lowered the score, and a change that doubles the cost only if by more than the seed range (0.029).

Model or change	Mean	Seed range
Seasonal naive	0.870	
LightGBM, one model per store	0.715	
Lift model (3 seeds)	0.713	0.683 to 0.734
Item encoder and readout (5 seeds)	0.699	0.669 to 0.724
World model (3 seeds)	0.707	0.679 to 0.750
World model + normalised roll-forward (3 seeds)	0.656	0.646 to 0.675
+ average of three fits (1 run)	0.665	
+ weekday as seven inputs (seed 1)	0.653	
+ weekday and event kinds (seed 1)	0.646	
+ 60,000 training steps (seed 1)	0.642	
+ item level against past-year level (seed 1)	0.680	

Items with no recent sales. On M5 the error at the chain total followed the error in the overall level (correlation 0.87 over 50 runs of every model on the four windows, Figure 22a). Grouping April's item-stores by their sales in March (Figure 22b), both models were within 6% of sales in every group that sold in March, and the world model was closer on 50.6% to 54.2% of item-stores in each. On item-stores that sold nothing in March, which sold 14,102 units in April, the world model forecast 62% too much and LightGBM 5% too little: the world model's weakest group. A first grouping by sales in the scored window itself had shown LightGBM's under-forecast growing with the sales rate, to 18% on the fastest items; that pattern came from grouping by the outcome being scored and is not reported as a finding.

C.1 What the first version tried and dropped

A lift-model seed that broke on five series. In the April M5 final window, lift-model seed 2 scored 2.603 over all levels against 0.566 and 0.570 for seeds 1 and 3, and forecast 1,413,724 units against 1,183,626 sold. Five item-store series hold all of the excess, each with its first price cut in a year inside the window; the seed read the cuts as lifts up to the model's bound. Item by store, weighted by

Table 26: Every change to a model setting with saved runs, scored against the configuration it changed; most were lower in error in at most half their comparisons, including some that were kept. A comparison is two configurations with the same data, windows and every other setting, scored on the windows and seeds both ran (overall WRMSSE, development windows, from the audit of the saved runs). Comparisons: the number of such pairs (categories, data sets or base configurations); lower: how many the change won. With it and without: mean WRMSSE over the pairs. Kept: whether the change is in the models of the main text. Loss changes are in Table 4; averaging fits, the stock term and runs without saved models are in the text and Table 27. Built by `scripts/change_ledger.py`.

Change	Tested on	Comparisons	Lower	With it	Without	Kept
Item level from the last 8 periods, against the whole history window	Dominick's	36	17	0.799	0.804	Yes (Dominick's)
Normalised roll-forward	Dominick's, M5	19	10	0.845	0.832	Yes (M5, Dominick's)
Own forecast fed into the roll-forward	Dominick's	18	4	0.910	0.816	No
Item level from the last 4 periods, against the whole history window	Dominick's	16	12	0.938	0.987	No
Item level from the last 8 periods, against the last 4	Dominick's	16	8	0.966	0.938	No
Brand switching	Dominick's	12	7	0.797	0.832	Yes (Dominick's)
Both levels as inputs, scaled by the 26-week level (against the 8-week level)	Dominick's	6	2	0.946	0.818	No
Level window set by the item's role	Dominick's	6	1	1.125	0.884	No
Roll-forward through the plan (world model against lift model)	Dominick's, Frat, M5	6	2	0.504	0.504	Yes (world model)
Plan split into regular and promoted parts	Dominick's, Frat	5	1	0.538	0.520	No
More training steps	Dominick's, Frat, M5	3	1	0.844	0.578	No
Plan of earlier periods as inputs	Frat	2	0	0.414	0.407	No
Year-ago sales as an input	Frat	2	0	0.413	0.409	No
Event kinds as inputs	M5	1	1	0.646	0.653	No
Longer history and training period	Frat	1	0	0.413	0.406	No
New-item start from similar items	Frat	1	1	0.461	0.594	New-item test only
Past-year level	M5	1	0	0.680	0.646	No
Readout dropout and weight decay	Frat	1	0	0.409	0.406	Yes (Frat lift model)
Recorded base price as the regular price	Frat	1	0	0.434	0.406	No
Regular price held for more periods	Dominick's	1	1	0.674	0.676	No
Unknown-item labels in training	Frat	1	1	0.459	0.461	No
Weekday and event kinds as inputs	M5	1	1	0.646	0.646	No
Weekday as seven inputs	M5	1	0	0.653	0.646	No

dollar sales, the failure is invisible (0.834); at the totals it dominates. The world model did not fail on these series in any seed.

Frat holiday weeks. On the Frat development windows the world model, lift model and LightGBM were level (0.406 to 0.421). On the final 8 weeks the world model fell to 0.681 against 0.549 for LightGBM. The final weeks are the holiday period, which the development windows did not contain (Figure 23); the world model's 8-week total is right (forecast against sales 1.000) but it is 15% to 17% under in the two December peak weeks. LightGBM, with week-of-year and month inputs and the previous holiday season in its training data, follows that shape more closely.

TimesFM's default output. TimesFM's default point forecast is the median. For an item that sells nothing on most days the median is near zero while the mean is not, so the median forecast is 27% to 31% below M5 totals in every month. The mean output, used throughout, does not have the problem.

Table 27: Results in the record not shown elsewhere in the paper, and configurations never run on full data. Overall WRMSSE on the development windows, lower is better; "against" gives the configuration it changed. Frat: five seeds; analgesics: three seeds; other Dominick's rows: one seed. Values transcribed from `results/README.md`.

Run	Where	Result
Item encoder and readout, whole stores drawn in training	Frat	0.430 against 0.424
Item encoder and readout, 10,000 steps	Frat	0.432 against 0.424
Lever step with the latent loss alone	Frat	0.428 against 0.429
Lift model, 52 weeks of history	Frat	0.413 against 0.406
Lift model, plan of the four weeks before and the same week last year	Frat	0.418 against 0.406
LightGBM with the seasonal calendar and promotion crowding	Frat	0.426 against 0.421
Lift model with validation stop, totals term, dropout and weight decay together	Frat	0.445 against 0.406
Lift model, latent pretraining, encoder at a tenth of the rate, latent loss kept on	Frat	0.415 against 0.406
World model, long latent pretraining, with validation stop and totals term	Frat	0.458 against 0.408
LightGBM, average of five fits	Frat	0.417 against 0.421
Lift model, average of five fits	Frat	0.400 against 0.406; the best Frat result of any single model family, not carried to the final tests (the record gives no reason)
Lift model, all training fixes with long training	Analgesics	0.716 against 0.676
World model with the totals term, and with the best Frat recipe	Analgesics	0.732 and 0.729 against 0.667
World model, latent check, VICReg penalty and near-week weighting	Analgesics	0.685 against 0.667
World model with switching, three fits averaged, without the normalised roll-forward	6 categories	Bottled juice 1.170 against 1.066
World model, full-length training, with seasonal inputs, and with a dollar-weighted loss	4 categories	Configured (job `dominicks_seasonal_long`); not run on full data
World model, unit-volume weighted loss	8 categories	Configured (job `dominicks_units_loss`); not run on full data. On a six-store sample (lift model, 800 steps) it cut the snack-cracker over-forecast from 1.63 to 1.39

Table 28: On M5 the best model depends on the level scored and on the month. WRMSSE on the four 28-day windows (lower is better), mean over seeds; seeds per window in the second column. All: the 12 levels together; Total: the chain total; I×S: item by store. The lowest value in each column is in bold. TimesFM with the plan has no recorded total. The lift model's April mean includes one failed seed (2.603 overall; §10).

		29 Feb to 27 Mar			28 Mar to 24 Apr			25 Apr to 22 May (practice)			23 May to 19 Jun (test)		
Model	**Seeds**	All	Total	I×S	All	Total	I×S	All	Total	I×S	All	Total	I×S
World model	3/3/3/3	**0.642**	**0.398**	0.851	**0.555**	**0.283**	0.833	0.656	0.535	0.834	0.638	0.379	0.897
Lift model	3/3/3/3	0.664	0.427	0.846	1.246	0.964	0.834	0.713	0.623	0.832	0.663	0.368	0.896
LightGBM	3/3/1/3	0.709	0.578	0.873	0.557	0.351	0.846	0.715	0.659	0.847	**0.547**	**0.247**	0.905
Graph model	3/3/3/3	0.778	0.621	0.878	0.644	0.493	0.845	**0.641**	**0.455**	0.851	0.673	0.412	0.909
TimesFM	1/1/1/1	0.854	0.830	**0.836**	0.819	0.786	**0.828**	0.896	0.930	**0.828**	0.729	0.558	0.885
TimesFM with the plan	1/1/1/1	0.858	–	0.837	0.772	–	0.829	0.929	–	0.830	0.719	–	**0.883**
Seasonal naive	1/1/1/1	0.889	0.658	1.121	0.923	0.668	1.126	0.870	0.751	1.100	0.847	0.560	1.176

Table 29: On the promotion-driven weekly data, models that see the plan lead, and which one leads changes by category. WRMSSE on the final 8 weeks (lower is better). World model, lift model and LightGBM: mean of three seeds, settings fixed on the development windows. TimesFM and TimesFM with the plan: one run each, added after the final tests. The lift model was run on Frat only. The lowest value in each row and group is in bold.

	All levels					Item by store				
Data	World	Lift	LightGBM	TimesFM	TimesFM+plan	World	Lift	LightGBM	TimesFM	TimesFM+plan
Frat	0.681	0.714	**0.549**	0.965	0.686	0.665	0.677	**0.608**	0.827	0.641
Dominick's										
Cheese	0.897	–	0.708	0.829	**0.545**	0.715	–	0.702	0.792	**0.699**
Snack crackers	0.448	–	**0.412**	0.468	0.475	0.608	–	**0.596**	0.640	0.614
Soft drinks	**0.830**	–	0.989	1.070	0.974	0.734	–	**0.652**	0.895	0.780
Bottled juice	0.987	–	0.994	**0.939**	0.990	0.816	–	**0.750**	0.782	0.754
Analgesics	**0.867**	–	0.978	1.008	0.889	**0.713**	–	0.714	0.790	0.742
Frozen entrees	0.485	–	**0.478**	0.563	0.566	0.557	–	**0.529**	0.682	0.599

Table 30: On the single final windows the ensemble is lowest or within 0.02 of the lowest item by store on every data set, and the order at the total changes from data set to data set. WRMSSE on the final window of each data set (lower is better; on M5 the official test month). One fit per trained model, so the world model and LightGBM differ slightly from the three-seed means of Tables 28 and 29. Bold: lowest in the group.

	Item by store							Total						
Data	Ens.	World	Lift	LGB	TFM+p	Chr.	TFT	Ens.	World	Lift	LGB	TFM+p	Chr.	TFT
Cheese	**0.675**	0.712	0.756	0.699	0.704	0.807	0.731	0.753	1.064	0.997	0.768	0.531	1.009	**0.522**
Snack crackers	**0.585**	0.610	0.620	0.603	0.624	0.661	0.621	0.334	0.484	0.302	0.317	0.362	0.327	**0.219**
Soft drinks	0.674	0.709	0.772	**0.655**	0.780	0.912	0.849	1.065	**0.891**	0.920	1.238	1.140	1.385	1.228
Bottled juice	0.760	0.849	0.807	0.756	**0.754**	0.785	0.775	1.111	1.019	1.190	1.171	1.172	1.046	**1.015**
Analgesics	**0.710**	0.712	0.714	0.716	0.742	0.802	0.770	1.236	1.186	1.280	1.407	1.149	1.454	**0.878**
Frozen entrees	0.543	0.574	0.563	**0.530**	0.599	0.692	0.557	0.454	0.423	**0.404**	0.438	0.537	0.536	0.414
Frat	**0.604**	0.671	0.662	0.613	0.644	0.856	0.702	0.586	0.681	0.698	**0.523**	0.677	1.222	0.818
M5	**0.881**	0.896	0.896	0.905	0.883	0.899	–	**0.240**	0.363	0.424	0.245	0.556	1.007	–

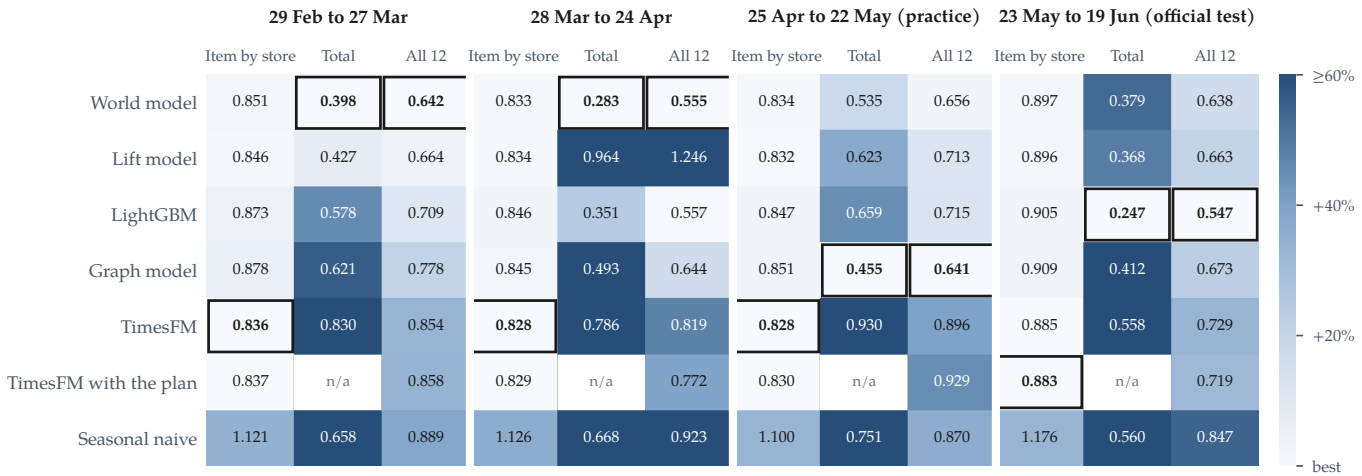


Figure 20: On the M5 final windows, TimesFM from history alone had the lowest item-by-store error of the single models in every month; at the total and over all levels the plan-aware models did, and which one changed by month. WRMSSE of every single model, four windows, three levels; cell shade: how far above the lowest value in its column (white: the lowest, outlined). Trained models: mean of three seeds. The lift model's April mean includes one failed seed (§C.1).

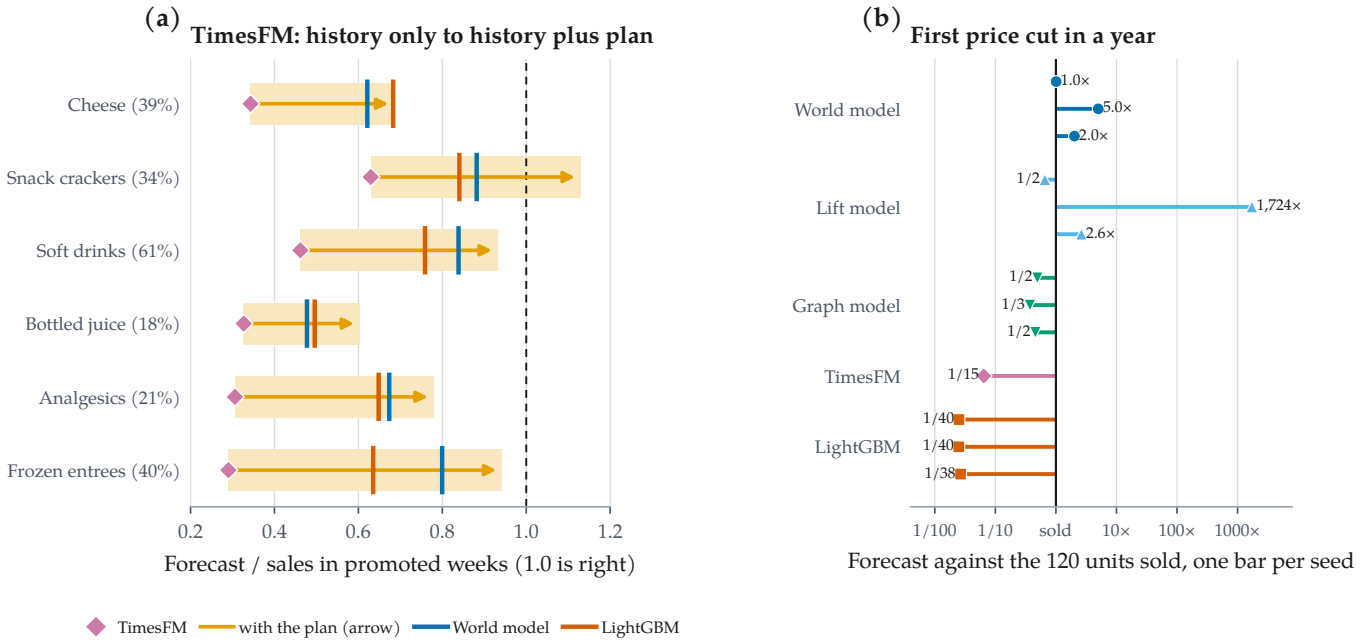


Figure 21: Given the plan, TimesFM closes most of its promoted-week shortfall; for a price cut history has not shown, plan-aware models respond and history-only models do not. **(a)** Forecast against sales in the promoted weeks of the final 8 weeks, by Dominick's category (promoted weeks' share of the category's units in brackets): TimesFM from history alone (diamond) to TimesFM with the plan (arrow head), with the world model and LightGBM (ticks). **(b)** HOUSEHOLD_1_146 in the four M5 stores where it had its first price cut in a year (5.98 to 4.50 for one week) inside the April window: each seed's forecast over the 28 days against the 120 units sold, log scale.

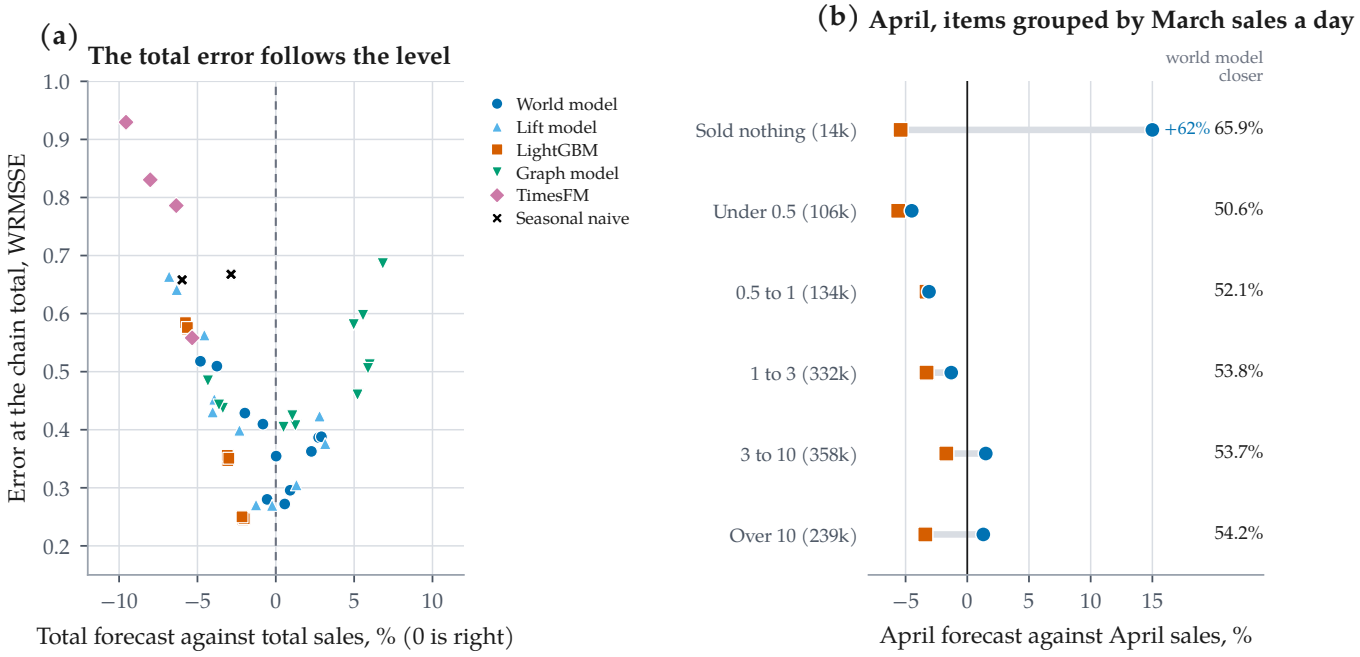


Figure 22: On M5 the error at the total followed the error in the level, and the world model over-forecast item-stores that sold nothing the month before. **(a)** Total forecast against total sales and WRMSSE at the chain total, every model and seed on the four M5 windows (the failed April lift seed left out). **(b)** April forecast against April sales by the item-store's sales a day in March (April units in brackets), world model and LightGBM, three seeds averaged; right: the share of item-stores where the world model was closer.

Table 31: Every change and comparison model that did not become a default. Overall WRMSSE unless stated; "against" gives the default it was compared with on the same window. dev.: development windows before the final 8 weeks. Practice: the M5 window 25 April to 22 May 2016. Each row is a section of `results/README.md` with its run folders.

Change or model	Where tested	Result
Twice the training steps	Dominick's (6) and Frat, dev., 3 seeds	Better in 2, worse in 3, level or mixed in 2
Weekday as seven inputs	M5 practice, seed 1	0.653 against 0.646
Event kinds (with weekday)	M5 practice, seed 1	0.646 against 0.646
60,000 training steps	M5 practice, seed 1	0.642 against 0.646; gain below seed range 0.029
Item level against past-year level	M5 practice, seed 1	0.680 against 0.646
Totals term in the loss	Dominick's analgesics, dev.	0.819 against 0.683; total 8% under
Negative binomial likelihood	Frat, analgesics, dev.	Level
Plan split, latent check, near-week weighting	Frat, dev., 5 seeds	0.417 to 0.427 against 0.408
Own forecast fed into the roll-forward	Dominick's (6), dev.	Worse overall in 4 of 6
Item role labels	Dominick's (6), dev.	No effect on LightGBM; snack crackers 2.422 for the world model
Averaging three fits	M5 practice, seed 1	0.665 against 0.656 (kept on Dominick's)
Graph model over product attributes	M5, four windows, 3 seeds	Level with LightGBM item by store
TimesFM, median output	M5, four windows	Overall 1.71 to 1.88; totals 27% to 31% under
Lift model (no roll-forward)	M5, four windows, 3 seeds	One seed failed in April (2.603)

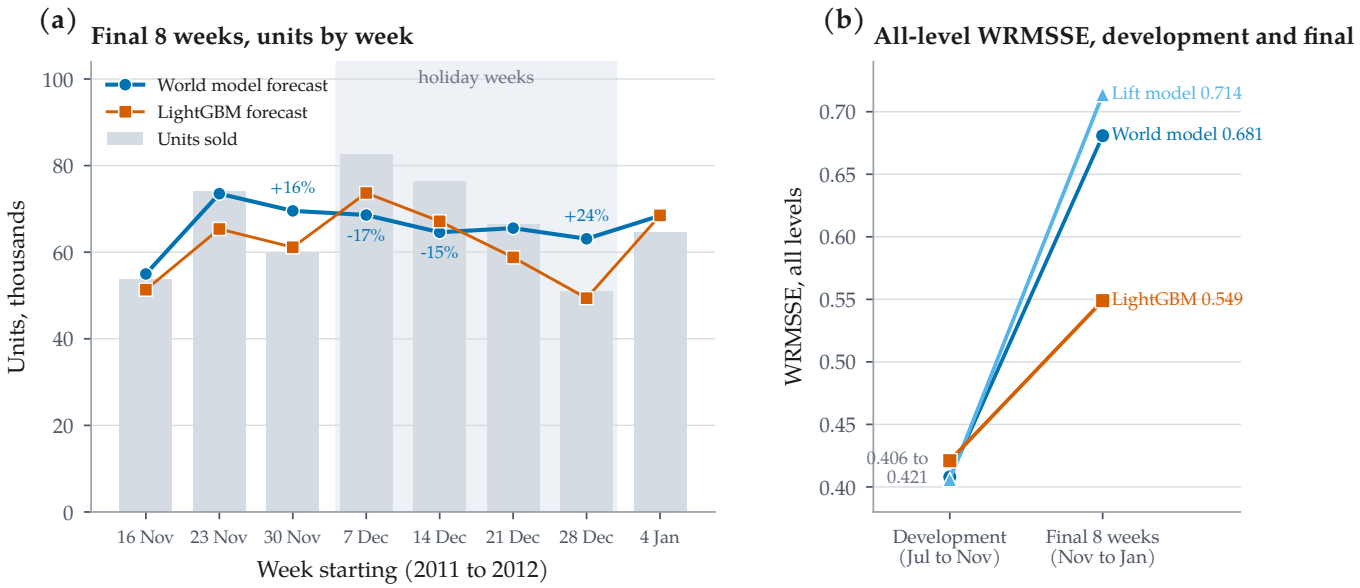


Figure 23: On the Frat final window the world model got the 8-week total right and missed the holiday shape, which the development windows did not contain. **(a)** Units sold each week of the final window over all item-stores (bars), and the weekly forecasts of the world model and LightGBM, three seeds averaged; labels: the world model's largest weekly misses. **(b)** All-level WRMSSE on the development windows and on the final 8 weeks.